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Study on vortex-induced multimodal coupled vibration of arch bridge suspenders

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ABSTRACT

The objective of this paper was to investigate the 2-DOF suspender response under the condition of vortex-induced force and reveal the multimodal coupling mechanism. Firstly, an elastically mounted segmental model of suspender was established. Wind tunnel experiments were conducted to obtain the time history displacement signals in cross-flow and in-line directions, the empirical parameters were obtained by identify those signals. The nonlinear partial differential motion equation of suspender with two degrees of freedom was established based on Hamilton's principle, which considered the dynamic strain, bending stiffness and vortex-induced force of suspender. The Galerkin method was used to discretize the nonlinear partial differential equations into nonlinear motion governing equations, which considered the first three order modes. A numerical method was used to solve the motion governing equation, the time history displacement curve, phase diagram, amplitude spectrum and Poincare map of the multi-modal responses in the cross-flow and in-line directions were obtained. With the weak coupling effect of nonlinear stiffness and vortex-induced force, the cross-flow and in-line responses demonstrate period doubling bifurcation, quasi-periodic vibration, period doubling vibration, crises and transient chaotic phenomena. However, with the strong coupling effect, the crises and transient chaos in response amplitude disappear. More importantly, planar/non-planar multimodal internal resonance were found in this work. With the weak multimodal coupling effect in suspender response, the low-order modal response is dominant; as the multimodal coupling effect increases, the high-order modal response becomes predominant. In addition, the system energy not only transfers from high-order modal response to low-order modal response, but also the system energy transfer from low-order modal response to high-order modal response. These conclusions are useful for the design and practical engineering of suspenders, which have high theoretical research value in understanding the mechanism of multimodal coupling effects on suspender response.

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1. Introduction

The slender rod with a large slenderness ratio fixed at both ends that has been extensively studied is the sagged flexible cable, with stay cables and conductors being the primary research objects. Rega [1] has provided a comprehensive review of theoretical and experimental studies on the nonlinear vibration of sagged flexible cables. However, there are also vertically installed slender poles in engineering applications, such as suspenders and offshore risers [2,3].

In practical engineering, Feng [4] observed a second-order modal response in the amplitude spectrum of the suspender response. The experimental results of Sun [5] also demonstrate the existence of multimodal responses in slender cables. In the vertical direction, due to the Boundary layer effect of the fluid, the wind speed will increase exponentially with the increase of height, that is, the wind velocity may exist at different heights of the structure [6,7]. Therefore, vertically installed suspenders are prone to multimodal coupled vibration under vortex shedding excitation, which in turn leads to fatigue and failure of the suspenders. Based on this, the research on vortex induced multi-mode coupled vibration of vertical members has megaproject significance and theoretical value.

Qu and Metrikine [8,9] established a new cylindrical structure wake oscillator model and studied the effects of cross-flow and in-line coupling on the wake oscillator model, respectively. In addition, Postnikov [10] used CFD to establish a circular cross-section model with elastic support and provided a new 2DOF vortex wake model. In fact, it is very difficult to solve the analytical expression of multi degree of freedom coupling [11], only semi analytical and semi numerical solutions can be given at present. Then, the coupling effect of one direction response on another direction response is harmonic transformed to obtain the polynomial of external excitation combination. This work uses a van der Pol oscillator combined with external excitation to form a wake oscillator model, then identifies the parameters of the cross-flow and in-line responses to obtain the vortex-induced force.

The focus of this work is on the suspender of a long-span arch bridge, which has a maximum slenderness ratio of 800 and shares some vibration characteristics with stay cables. As subjected to external excitation, geometric nonlinearity and nonlinear coupling effects, slender rods with large slenderness ratios exhibit complex nonlinear behavior [12,13]. For example, Nayfeh [14] found that sagged cables under the effect of 1:2 internal resonance caused by secondary nonlinear coupling exhibit quasi-periodic and chaotic motion. Zhang [15] established a 3-degree of freedom iced transmission lines model under aerodynamic conditions, which exhibited one to multiple periods of chaotic motion due to in-plane parametric resonance and out-of-plane super-harmonic resonance. In this work, the suspender response also exhibits chaotic motion under the effect of near principal resonance, nonlinear effects and non-planar modal coupling effect caused by dynamic tension. Additionally, high-order modes of slender rods significantly contribute to the system response, as demonstrated by Bao [16] and Alfossail [17]. Feng [4] observed a harmonic disturbance twice the dominant frequency in the amplitude spectrum of the measured displacement response of the suspender, which may be caused by the 1:2 internal resonance of the second-order modal response. Compared to stay cables and transmission lines, the suspender has a much higher tension, resulting in stronger nonlinear coupling characteristics caused by dynamic tension during vibration process. This work aims to investigate the vortex-induced nonlinear vibration characteristics of suspenders from the perspective of multimodal coupling, focusing on the effects of non-planar modal interactions and multimodal interactions.

In the field of multimodal interactions, Cui [18] established a multimodal interactions model of conductor and discussed the coupling mechanism of the first two modes. This work also verified the occurrence of different energy exchanges between different modes, which is relevant to the work of the multimodal interaction mechanism of the suspender. However, as a load-bearing component, the suspender has greater tension and stiffness. In this study, the first three modes coupling of the suspender is considered. Srinil [19] analyzed the vortex-induced vibration of an elastic cable considered modal interaction and found that the simultaneous existence of primary and internal resonances resulted in both primary and secondary locking regimes in the structural response. The numerical results of Srinil [19] and the experimental results of Sun [5] demonstrate the existence of energy exchange between different order modes. Interestingly, the law of energy transfer in the system [5,19] is from high-order modal responses to low-order modal responses, which has also been verified by this work. In addition, this work found that system energy can also be transferred from low-order modal response to high-order modal response. The law of energy transfer is determined by system nonlinearity, coupling strength and the internal resonance effect. To facilitate the identification of different types of modal coupling effects, this work classifies them into non-planar modal coupling effect, planar multimodal coupling effect and non-planar multimodal coupling effect. Non-planar multimodal coupling effect can be subdivided into non-planar same-order modal coupling effect and non-planar different-order modal coupling effect. In this work, the non-planar modal coupling effect, planar multimodal coupling effect and non-planar multimodal coupling effect were founded in the suspender response considered multi-modal interactions. For the two-degree-of-freedom response considering only the first-order mode, only non-planar modal coupling effect in the system response, there is no multimodal interaction [20]. For a single degree of freedom response considering high-order modes [21], only planar multimodal coupling in the system response, and there is no non-planar modal interaction and non-planar multimodal interaction.

Modal interaction is closely related to internal resonance in the system response, which typically occurs in multiple-degree-of-freedom systems [22]. The energy transfer in the system between the different degrees of freedom was mainly reflected in the response amplitudes slowly changing over time like amplitude modulation waves, with the peak of one degree of freedom response amplitude corresponding to the other degree of freedom response amplitude [23,24]. When the ratio of a pair natural frequencies in a multi-degree-of-freedom system is an integer, there is significant energy exchange between two of them, wherein the coupling effect was caused by the effect of internal resonance [25,26]. In this work, the ratio of natural frequencies of each mode in the cross-flow and in-line directions is a positive integer, which means that the energy exchange phenomenon between different modal responses was caused by the effect of internal resonance. In other words, the internal resonance corresponding to non-planar modal interaction is called non-planar modal internal resonance [27]; the internal resonance corresponding to planar multimodal interaction is called planar multimodal internal resonance; the internal resonance corresponding to non-planar multimodal coupling effect is called

non-planar multimodal internal resonance [28]. In this work, the system energy not only transfers from high-order modal response to low-order modal response, but there is also a trend of system energy transfer from low-order modal response to high-order modal response. Therefore, to distinguish between these two trends of energy transfer in system, the energy flow direction of the high-order modal response to the low-order modal response is referred to as the "high: low internal resonance effect", while the energy flow direction of the low-order modal response to the high-order modal response is referred to as the "low: high internal resonance effect". In summary, this work studied the multimodal coupling response of suspender, considering the vortex-induced force, dynamic tension and bending stiffness of suspender. This work is of great significance for the study of vortex induced vibration of suspenders, of course is also applicable to similar engineering structures. In addition, it also contributes to the understanding of high-order modal coupling effects in engineering structures.

This work is structured as follows: In Section 2, introduction of suspender experiment model and wind tunnel test, establishment of suspender theoretical model of arch bridge. In Section 3, the influence of the non-planar modal interaction on the system response was been analyzed. In addition, the influence of the multi-modal interactions of suspenders on the dynamic response was been detailed discussed in Section 4. A detailed summary of results is described in Section 5.

2. The experiment model and theoretical model of suspender

2.1. Introduction to the experiment model

The experiment on the suspenders model was conducted at the wind tunnel laboratory of Central South University in Changsha, China. The wind tunnel experiment utilized an elastic suspension system, as depicted in Fig. 1(a), which consisted of an IL-300 laser displacement system, a segment model, 12 mechanical springs and other components, as shown in Fig. 1(b). During the vibration process, the cross-flow response signal of segmental model was collected by the laser displacement system (D1–3), while the in-line response signal was collected by the laser displacement system (D4–5). For this work, the time history displacement data of cross-flow direction were obtained from the average values of D1–3, the time history displacement data of in-line direction were obtained from the average values of D4 and D5.

The vortex vibration wind speed corresponding to the model vortex vibration was obtained through experiments. Figs. 2 and 3 show the experimental results obtained by applying an initial amplitude excitation in the cross-flow and in-line directions at vortex vibration wind speed. From the Figs. 2 and 3, it can be seen that the cross-flow response and the in-line response are mutually coupled; if the cross-flow and in-line responses are not coupled, then the transverse and downstream responses will decay uniformly to a stable state of vortex vibration under the combined effect of structural damping and aerodynamic damping; however, in Fig. 2, the decay process of response amplitude is not linear; the feature of beat vibration also appears in Fig. 3. In the amplitude spectrum of Figs. 2 and 3, in addition to the response corresponding to the dominant frequency, there is also a harmonic interference near the dominant frequency. In Fig. 2, it can be seen from the amplitude spectrum that the dominant frequency and amplitude of the experimental model in the cross-flow direction are 3.800 Hz/2.589 mm. Besides, there is a harmonic resonant response with an amplitude of 0.481 mm near the frequency of 3.760 Hz. In Fig. 3, it shows the phenomenon of frequency doubling, which experience an attenuation stage; the dominant frequency (3.800 Hz) in the in-line direction is consistent with that in the cross-flow direction, which the corresponding amplitude is 0.464 mm. The harmonic resonant with an amplitude of 0.409 mm near the frequency of 3.737 Hz, which is the coupling effect of cross-flow response.

For the response characteristics in Fig. 2, due to the small coupling effect of the in-line direction on the cross-flow direction, the van der Pol oscillators can effectively identify aerodynamic parameters. For the response characteristics in Fig. 3, due to the coupling effect of the cross-flow and in-line directions, the amplitude spectrum exhibits a frequency doubling response. In order to obtain the accurate aerodynamic parameters, the coupling effect is considered as harmonic interference, a van der Pol oscillator containing external

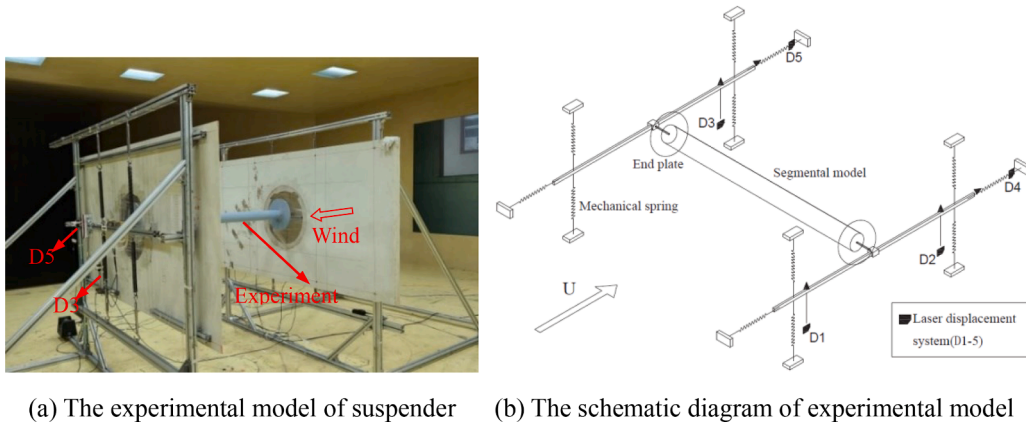


Fig. 1. The equipment installation diagram of experimental model. (a) The experimental model of suspender (b) The schematic diagram of experimental model.

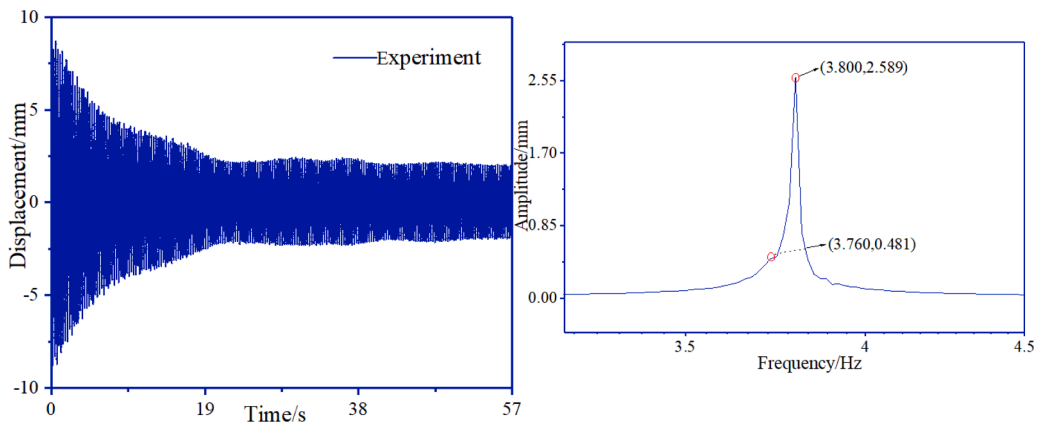


Fig. 2. The time history displacement curve and amplitude spectrum of cross-flow direction.

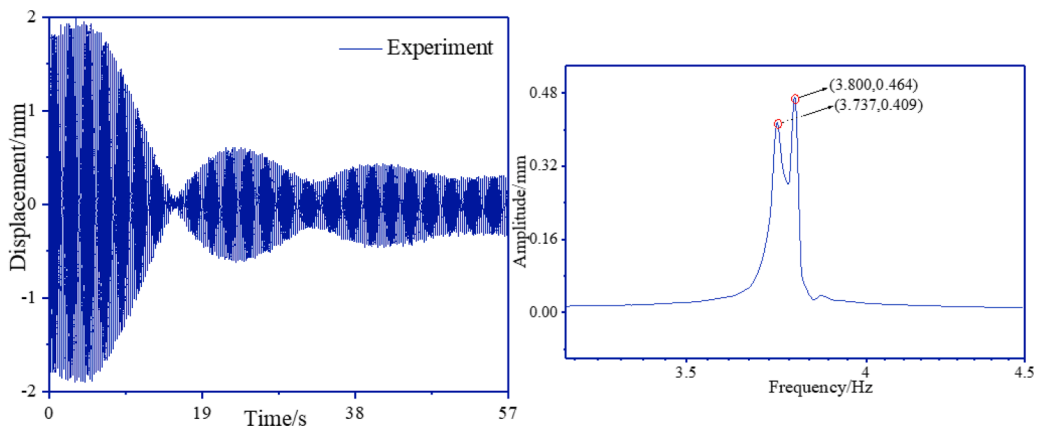


Fig. 3. The time history displacement curve and amplitude spectrum of in-line direction.

excitation terms was used to identify the VIV parameters in the cross flow and in-line directions [29]. The vortex induced characteristic parameters (the attenuation time, the ratio of initial excitation amplitude to vortex induced stability amplitude) in Figs. 2 and 3 were taken into the analytical expression in Appendix. A, the vortex induced parameters (a , b , p^* , a_1 , b_1 , p_1^*) in Table 1 can be obtained. Then, the frequency(ω_{d1}^2 , ω_{d2} , ω_{l1}^2 , ω_{l2}) can be obtained by converting the dominant frequency into circular frequency in Figs. 2 and 3. The identification results of empirical parameters were shown in Table 1, which the negative damping term, stiffness damping term and disturbance excitation term constitute the aerodynamic load of the oscillator model. It is worth noting that the aerodynamic load on the oscillator model represents the aerodynamic load on each meter of the segment model. Due to the focus of the work on the multimodal coupling effect of suspender response, the relevant aerodynamic load model has been simplified. However, this aerodynamic load can still reflect the vortex induced forces in both the cross-flow and in-line directions.

2.2. The theoretical model of arch bridge suspender

Fig. 4 is the suspender model of arch bridge. In Fig. 4(a), $u(s,t)$, $v(s,t)$, $w(s,t)$ are the displacements measured from the dynamic equilibrium configuration in the x-axis, y-axis and z-axis directions respectively, U is the horizontal wind load. The initial static equilibrium configuration is ds , and the dynamic equilibrium configuration is ds_1 . It is considered that the wind load is along the y-axis

Table 1
The identification results of empirical parameters.

Parameter	Symbol(cross-flow/in-line)	Value(cross-flow/in-line)
nature frequency	$\omega_{d1}^2/\omega_{l1}^2$	570.068/570.068
negative damping	a/a_1	-0.055/-0.049
stiffness damping	b/b_1	0.084/1.608
disturbance excitation	p^*/p_1^*	7.306/9.755
disturbance frequency	ω_{d2}/ω_{l2}	23.625/23.480

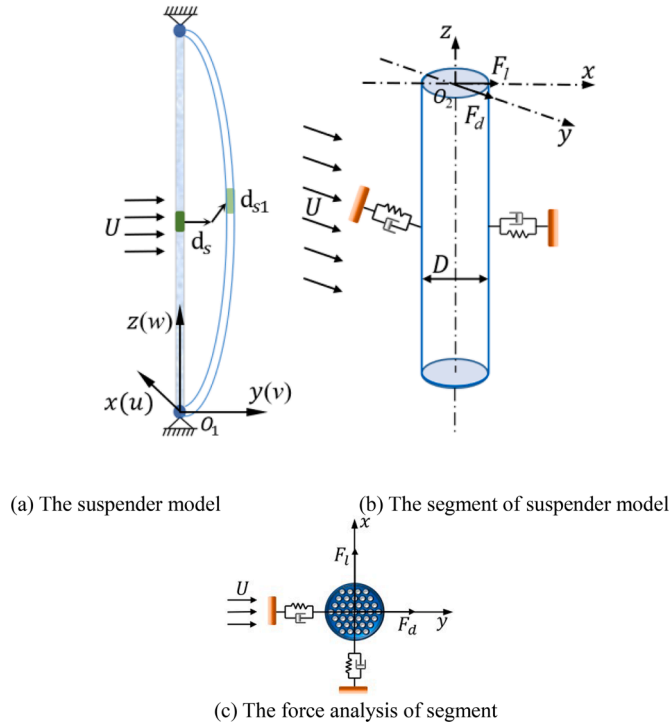


Fig. 4. The suspender model of arch bridge. (a) The suspender model (b) The segment of suspender model (c) The force analysis of segment.

direction. Fig. 4(b) is the suspender model. Taking the micro-segment ds as the research objects of this work; the D is the diameter of segment model. Fig. 4(c) depicts the force analysis model of the arch bridge suspender cross section. The F_d is the aerodynamic load along the wind direction, the F_l is the aerodynamic load perpendicular to the wind direction.

By incorporating the aerodynamic load in Table 1 into the model in Fig. 4(c), the resulting formula is as follows:

$$\begin{aligned} F_d &= a^* \frac{\partial}{\partial t} u + b^* u^2 \frac{\partial}{\partial t} u - p \cos(\omega_d t) \\ F_l &= a_1^* \frac{\partial}{\partial t} v + b_1^* v^2 \frac{\partial}{\partial t} v - p_1 \cos(\omega_l t) \end{aligned} \quad (1a-b)$$

Eq. (1) only represents the aerodynamic load on each meter of the suspender in the transverse and downstream directions, which is obtained by converting the scaled model of the experimental section with the actual size of the suspender.

In order to analyze the multimodal coupling effect of suspender, the following assumptions were made in this work:

- (1) Ignoring the axial stiffness and shearing stiffness of suspender.
- (2) Considering the 2-DOF model of vertical-transverse bending of suspender, the axial deformation of suspender is ignored.
- (3) The stress-strain relationship of suspender follows Hooke's law.
- (4) Since the tension of the suspender is far greater than the gravity of the suspender itself, it is assumed that the vertically tension of suspender is equal everywhere under static conditions.

In Fig. 4(a), during the vibration process of suspender (from d_s to d_{s1}), the strain is expressed as:

$$\varepsilon = (d_{s1}^2 - d_s^2) / 2d_s^2 \quad (2)$$

As a result of the axial z -direction deformation of suspender is negligible compared with the cross-flow and in-line deformation, the Eq. (2) is rewritten as follows:

$$\varepsilon = \frac{1}{2} \left(\frac{\partial}{\partial s} u \right)^2 + \frac{1}{2} \left(\frac{\partial}{\partial s} v \right)^2 + \frac{\partial}{\partial s} u \frac{\partial}{\partial s} x + \frac{\partial}{\partial s} v \frac{\partial}{\partial s} y \quad (3)$$

According to Hamiltonian principle:

$$\delta H = \int_{t_1}^{t_2} (\delta K - \delta \Pi + \delta W) dt = 0 \quad (4)$$

In Eq. (4), δK is kinetic energy, $\delta \Pi$ is potential energy and δW is the other forms of energy.

$$K = -\frac{1}{2}m \int_0^l \left(u \frac{\partial^2}{\partial t^2} u + v \frac{\partial^2}{\partial t^2} v \right) \quad (5)$$

$$\Pi = \int_0^l \left(T\epsilon + \frac{ES}{2}\epsilon^2 \right) ds + \frac{EI}{2} \int_0^l \left[\left(\frac{\partial^2}{\partial s^2} u \right)^2 + \left(\frac{\partial^2}{\partial s^2} v \right)^2 \right] ds \quad (6)$$

$$W = \int_0^l \left(F_{d,l} - \mu_{d,l} \frac{\partial}{\partial t} u, v \right) ds \quad (7)$$

In Eqs. (5)-(7), m is the mass per unit length of suspender, T is the tension of suspender, E is the elastic modulus of suspender, I is the sectional moment of inertia, μ_d and μ_l are the damping coefficients in the in-line and cross-flow directions, respectively.

Let H be the vertical component of static tension T . As a result of the cross-flow (along the x direction) VIV contributes the most to the suspender vibration, then $ds \approx dx$, $H \approx T$. Moreover, the static configuration of suspender is a straight line in vertical direction, the initial deflection in the cross-flow and in-line directions is 0, that is, $dy/dx=0$. The suspender response produces geometric nonlinearity in the cross-flow and in-line directions. As a result of the displacement in the cross-flow and in-line directions is far less than the length of suspender, the influence of this geometric nonlinearity on the structural response is ignored. Then the nonlinear partial differential equation of motion in the cross-flow direction is:

$$H \frac{\partial^2}{\partial x^2} u + \frac{1}{2} ES \frac{\partial^2}{\partial x^2} u \int_0^l \frac{1}{2} \left[\left(\frac{\partial}{\partial x} u \right)^2 + \left(\frac{\partial}{\partial x} v \right)^2 \right] dx - EI \frac{\partial^4}{\partial x^4} u = m \frac{\partial^2}{\partial t^2} u - 2F_d + 2\mu_d \frac{\partial}{\partial t} u \quad (8)$$

Similarly, the nonlinear partial differential equation of motion in the in-line direction can be obtained:

$$H \frac{\partial^2}{\partial x^2} v + \frac{1}{2} ES \frac{\partial^2}{\partial x^2} v \int_0^l \frac{1}{2} \left[\left(\frac{\partial}{\partial x} u \right)^2 + \left(\frac{\partial}{\partial x} v \right)^2 \right] dx - EI \frac{\partial^4}{\partial x^4} v = m \frac{\partial^2}{\partial t^2} v - 2F_l + 2\mu_l \frac{\partial}{\partial t} v \quad (9)$$

For the nonlinear dynamic analysis, the modal truncation method is adopted to truncate the third order mode:

$$\begin{aligned} u(x, t) &= f_1(x)q_1(t) + f_2(x)q_2(t) + f_3(x)q_3(t) \\ v(x, t) &= \vartheta_1(x)\psi_1(t) + \vartheta_2(x)\psi_2(t) + \vartheta_3(x)\psi_3(t) \end{aligned} \quad (10a-b)$$

As a result of the main purpose of this work is to study the effect of multimodal interactions on suspender response, it is assumed that the harmonic interference(coupling effect) is uniformly distributed along the height direction, which is shown in Eq. (11)(a). Due to the physical characteristics of suspender in the cross-flow direction and the in-line direction are consistent, the consistent modal function is used in this work. The modal function in Eq. (10) is shown in Eq. (11)(b).

$$\begin{aligned} p_{d,l}^* &= f_0(x)p_{d,l}, f_0(x) = 1; \\ f_i &= \vartheta_i = \sin(i\pi x/l) (i = 1, 2, 3/i = 1, 3, 5); \end{aligned} \quad (11a-b)$$

For the nonlinear dynamic analysis, based on the Galerkin method, substituting Eqs. (10) and (11) into Eqs. (8) and (9) yielded:

$$\begin{aligned} &\ddot{q}_1 + \omega_1^2 q_1 + c_{1l} \dot{q}_1^3 / 2 + c_{2l} q_1 \dot{q}_2^2 / 2 + c_{3l} q_1 \dot{q}_3^2 / 2 + c_{4l} q_2 \dot{q}_1^2 + c_{5l} q_3 \dot{q}_1^2 + c_{6l} q_1 \dot{q}_2 \dot{q}_3 + c_{7l} q_1 \dot{\psi}_1^2 / 2 \\ &+ c_{8l} q_1 \dot{\psi}_2^2 / 2 + c_{9l} q_1 \dot{\psi}_3^2 / 2 + c_{10l} q_1 \dot{\psi}_1 \dot{\psi}_2 + c_{11l} q_1 \dot{\psi}_1 \dot{\psi}_3 + c_{12l} q_1 \dot{\psi}_2 \dot{\psi}_3 + c_{13l} \dot{q}_1 + c_{14l} \dot{q}_1 \dot{q}_1^2 + c_{15l} \dot{q}_2 \dot{q}_2^2 \\ &+ c_{16l} \dot{q}_3 \dot{q}_3^2 + c_{17l} (\dot{q}_1^2 \dot{q}_2 + 2\dot{q}_1 \dot{q}_1 \dot{q}_2) + c_{18l} (\dot{q}_1^2 \dot{q}_3 + 2\dot{q}_1 \dot{q}_1 \dot{q}_3) + c_{19l} (\dot{q}_2^2 \dot{q}_1 + 2\dot{q}_2 \dot{q}_2 \dot{q}_1) + c_{20l} (\dot{q}_2^2 \dot{q}_3 \\ &+ 2\dot{q}_2 \dot{q}_2 \dot{q}_3) + c_{21l} (\dot{q}_3^2 \dot{q}_2 + 2\dot{q}_3 \dot{q}_3 \dot{q}_2) + c_{22l} (\dot{q}_3^2 \dot{q}_1 + 2\dot{q}_3 \dot{q}_3 \dot{q}_1) + 2c_{23l} (\dot{q}_1 \dot{q}_2 \dot{q}_3 + q_1 \dot{q}_2 \dot{q}_3 + q_1 \dot{q}_2 \dot{q}_3) + c_{24l} p_{ci} = 0 \\ &\ddot{\psi}_1 + \omega_2^2 \psi_1 + d_{1l} \dot{\psi}_1^3 / 2 + d_{2l} \dot{\psi}_1 \dot{\psi}_2^2 / 2 + d_{3l} \dot{\psi}_1 \dot{\psi}_3^2 / 2 + d_{4l} \dot{\psi}_2 \dot{\psi}_1^2 + d_{5l} \dot{\psi}_3 \dot{\psi}_1^2 + d_{6l} \dot{\psi}_1 \dot{\psi}_2 \dot{\psi}_3 \\ &+ d_{7l} \dot{\psi}_1 \dot{q}_1^2 / 2 + d_{8l} \dot{\psi}_1 \dot{q}_2^2 / 2 + d_{9l} \dot{\psi}_1 \dot{q}_3^2 / 2 + d_{10l} \dot{\psi}_1 q_1 \dot{q}_2 + d_{11l} \dot{\psi}_1 q_1 \dot{q}_3 + d_{12l} \dot{\psi}_1 q_2 \dot{q}_3 + d_{13l} \dot{\psi}_1 + d_{14l} \dot{\psi}_1 \dot{\psi}_1^2 \\ &+ d_{15l} \dot{\psi}_2 \dot{\psi}_2^2 + d_{16l} \dot{\psi}_3 \dot{\psi}_3^2 + d_{17l} (\dot{\psi}_1^2 \dot{\psi}_2 + 2\dot{\psi}_1 \dot{\psi}_1 \dot{\psi}_2) + d_{18l} (\dot{\psi}_1^2 \dot{\psi}_3 + 2\dot{\psi}_1 \dot{\psi}_1 \dot{\psi}_3) + d_{19l} (\dot{\psi}_2^2 \dot{\psi}_1 + 2\dot{\psi}_2 \dot{\psi}_2 \dot{\psi}_1) \\ &+ d_{20l} (\dot{\psi}_2^2 \dot{\psi}_3 + 2\dot{\psi}_2 \dot{\psi}_2 \dot{\psi}_3) + d_{21l} (\dot{\psi}_3^2 \dot{\psi}_2 + 2\dot{\psi}_3 \dot{\psi}_3 \dot{\psi}_2) + d_{22l} (\dot{\psi}_3^2 \dot{\psi}_1 + 2\dot{\psi}_3 \dot{\psi}_3 \dot{\psi}_1) + 2d_{23l} (\dot{\psi}_1 \dot{\psi}_2 \dot{\psi}_3 \\ &+ \dot{\psi}_1 \dot{\psi}_2 \dot{\psi}_3 + \dot{\psi}_1 \dot{\psi}_2 \dot{\psi}_3) + d_{24l} p_{di} = 0 \end{aligned} \quad (12a-b)$$

$$\begin{aligned}
& \ddot{q}_2 + \omega_2^2 q_2 + c_{1j} \dot{q}_2^3 / 2 + c_{2j} q_2 \dot{q}_1^2 / 2 + c_{3j} q_2 \dot{q}_3^2 / 2 + c_{4j} q_1 \dot{q}_2^2 + c_{5j} q_1 q_2 q_3 + c_{6j} q_2^2 q_3 + c_{7j} q_2 \dot{\psi}_1^2 / 2 \\
& + c_{8j} q_2 \dot{\psi}_2^2 / 2 + c_{9j} q_2 \dot{\psi}_3^2 / 2 + c_{10j} q_2 \psi_1 \psi_2 + c_{11j} q_2 \psi_1 \psi_3 + c_{12j} q_2 \psi_2 \psi_3 + c_{13j} \dot{q}_1 + c_{14j} \dot{q}_1 \dot{q}_1^2 + c_{15j} \dot{q}_2 \dot{q}_2^2 \\
& + c_{16j} \dot{q}_3 \dot{q}_3^2 + c_{17j} (\dot{q}_1^2 \dot{q}_2 + 2 \dot{q}_1 q_1 \dot{q}_2) + c_{18j} (\dot{q}_1^2 \dot{q}_3 + 2 \dot{q}_1 q_1 \dot{q}_3) + c_{19j} (\dot{q}_2^2 \dot{q}_1 + 2 \dot{q}_2 q_2 \dot{q}_1) + c_{20j} (\dot{q}_2^2 \dot{q}_3 + 2 \dot{q}_2 q_2 \dot{q}_3) \\
& + c_{21j} (\dot{q}_3^2 \dot{q}_2 + 2 \dot{q}_3 q_3 \dot{q}_2) + c_{22j} (\dot{q}_3^2 \dot{q}_1 + 2 \dot{q}_3 q_3 \dot{q}_1) + c_{23j} (\dot{q}_1 \dot{q}_2 q_3 + q_1 \dot{q}_2 q_3 + q_1 q_2 q_3) + c_{24j} p_{ej} = 0 \\
& \ddot{\psi}_2 + \omega_2^2 \psi_2 + d_{1j} \dot{\psi}_2^3 / 2 + d_{2j} \psi_2 \dot{\psi}_1^2 / 2 + d_{3j} \psi_2 \dot{\psi}_3^2 / 2 + d_{4j} \psi_1 \dot{\psi}_2^2 + d_{5j} \psi_1 \dot{\psi}_2 \dot{\psi}_3 + d_{6j} \dot{\psi}_2^2 \dot{\psi}_3 \\
& + d_{7j} \psi_2 \dot{q}_1^2 / 2 + d_{8j} \psi_2 \dot{q}_2^2 / 2 + d_{9j} \psi_2 \dot{q}_3^2 / 2 + d_{10j} \psi_2 q_1 q_2 + d_{11j} \psi_2 q_1 q_3 + d_{12j} \psi_2 q_2 q_3 + d_{13j} \dot{\psi}_1 + d_{14j} \dot{\psi}_1 \dot{\psi}_1^2 \\
& + d_{15j} \dot{\psi}_2 \dot{\psi}_2^2 + d_{16j} \dot{\psi}_3 \dot{\psi}_3^2 + d_{17j} (\dot{\psi}_1^2 \dot{\psi}_2 + 2 \dot{\psi}_1 \psi_1 \dot{\psi}_2) + d_{18j} (\dot{\psi}_1^2 \dot{\psi}_3 + 2 \dot{\psi}_1 \psi_1 \dot{\psi}_3) + d_{19j} (\dot{\psi}_2^2 \dot{\psi}_1 + 2 \dot{\psi}_2 \psi_2 \dot{\psi}_1) \\
& + d_{20j} (\dot{\psi}_2^2 \dot{\psi}_3 + 2 \dot{\psi}_2 \psi_2 \dot{\psi}_3) + d_{21j} (\dot{\psi}_3^2 \dot{\psi}_2 + 2 \dot{\psi}_3 \psi_3 \dot{\psi}_2) + d_{22j} (\dot{\psi}_3^2 \dot{\psi}_1 + 2 \dot{\psi}_3 \psi_3 \dot{\psi}_1) + 2 d_{23j} (\dot{\psi}_1 \dot{\psi}_2 \dot{\psi}_3 \\
& + \psi_1 \dot{\psi}_2 \dot{\psi}_3 + \psi_1 \dot{\psi}_2 \dot{\psi}_3) + d_{24j} p_{dj} = 0
\end{aligned} \tag{12c-d}$$

$$\begin{aligned}
& \ddot{q}_3 + \omega_3^2 q_3 + c_{1k} \dot{q}_3^3 / 2 + c_{2k} q_3 \dot{q}_1^2 / 2 + c_{3k} q_3 \dot{q}_2^2 / 2 + c_{4k} q_1 \dot{q}_3^2 + c_{5k} q_2 \dot{q}_3^2 + c_{6k} q_1 q_2 q_3 + c_{7k} q_3 \dot{\psi}_1^2 / 2 \\
& + c_{8k} q_3 \dot{\psi}_2^2 / 2 + c_{9k} q_3 \dot{\psi}_3^2 / 2 + c_{10k} q_3 \psi_1 \psi_2 + c_{11k} q_3 \psi_1 \psi_3 + c_{12k} q_3 \psi_2 \psi_3 + c_{13k} \dot{q}_1 + c_{14k} \dot{q}_1 \dot{q}_1^2 + c_{15k} \dot{q}_2 \dot{q}_2^2 \\
& + c_{16k} \dot{q}_3 \dot{q}_3^2 + c_{17k} (\dot{q}_1^2 \dot{q}_2 + 2 \dot{q}_1 q_1 \dot{q}_2) + c_{18k} (\dot{q}_1^2 \dot{q}_3 + 2 \dot{q}_1 q_1 \dot{q}_3) + c_{19k} (\dot{q}_2^2 \dot{q}_1 + 2 \dot{q}_2 q_2 \dot{q}_1) + c_{20k} (\dot{q}_2^2 \dot{q}_3 \\
& + 2 \dot{q}_2 q_2 \dot{q}_3) + c_{21k} (\dot{q}_3^2 \dot{q}_2 + 2 \dot{q}_3 q_3 \dot{q}_2) + c_{22k} (\dot{q}_3^2 \dot{q}_1 + 2 \dot{q}_3 q_3 \dot{q}_1) + 2 c_{23k} (\dot{q}_1 \dot{q}_2 q_3 + q_1 \dot{q}_2 q_3 + q_1 q_2 q_3) + c_{24k} p_{ck} = 0 \\
& \ddot{\psi}_3 + \omega_3^2 \psi_3 + d_{1k} \dot{\psi}_3^3 / 2 + d_{2k} \psi_3 \dot{\psi}_1^2 / 2 + d_{3k} \psi_3 \dot{\psi}_2^2 / 2 + d_{4k} \psi_1 \dot{\psi}_3^2 + d_{5k} \psi_2 \dot{\psi}_3^2 + d_{6k} \psi_1 \dot{\psi}_2 \dot{\psi}_3 \\
& + d_{7k} \psi_3 \dot{q}_1^2 / 2 + d_{8k} \psi_3 \dot{q}_2^2 / 2 + d_{9k} \psi_3 \dot{q}_3^2 / 2 + d_{10k} \psi_3 q_1 q_2 + d_{11k} \psi_3 q_1 q_3 + d_{12k} \psi_3 q_2 q_3 + d_{13k} \dot{\psi}_1 + d_{14k} \dot{\psi}_1 \dot{\psi}_1^2 \\
& + d_{15k} \dot{\psi}_2 \dot{\psi}_2^2 + d_{16k} \dot{\psi}_3 \dot{\psi}_3^2 + d_{17k} (\dot{\psi}_1^2 \dot{\psi}_2 + 2 \dot{\psi}_1 \psi_1 \dot{\psi}_2) + d_{18k} (\dot{\psi}_1^2 \dot{\psi}_3 + 2 \dot{\psi}_1 \psi_1 \dot{\psi}_3) + d_{19k} (\dot{\psi}_2^2 \dot{\psi}_1 + 2 \dot{\psi}_2 \psi_2 \dot{\psi}_1) \\
& + d_{20k} (\dot{\psi}_2^2 \dot{\psi}_3 + 2 \dot{\psi}_2 \psi_2 \dot{\psi}_3) + d_{21k} (\dot{\psi}_3^2 \dot{\psi}_2 + 2 \dot{\psi}_3 \psi_3 \dot{\psi}_2) + d_{22k} (\dot{\psi}_3^2 \dot{\psi}_1 + 2 \dot{\psi}_3 \psi_3 \dot{\psi}_1) + 2 d_{23k} (\dot{\psi}_1 \dot{\psi}_2 \dot{\psi}_3 \\
& + \psi_1 \dot{\psi}_2 \dot{\psi}_3 + \psi_1 \dot{\psi}_2 \dot{\psi}_3) + d_{24k} p_{dk} = 0
\end{aligned} \tag{12e-f}$$

The parameters in Eq. (12) presented in Appendix. B. The physical parameters of the suspender are shown in Table 2:

In Fig. 5, it is the research ideas of this work. The VIVR of the 2-DOF elastic suspension suspender segment model was obtained through wind tunnel testing. The aerodynamic load of suspender was obtained by processing the vortex-induced response of experimental model. The two-degree-of-freedom partial differential motion equation of suspender was obtained by using Hamilton's principle, taking into account the dynamic tension and bending stiffness of suspender. The cross-flow and in-line multimodal motion governing equations, considering the coupling effect of the first two modes and first three modes, were obtained by using the modal superposition method and Galerkin discretization method. The motion governing equation was solved using the Runge-Kutta function. The solved data were processed to obtain time-history displacement curves, phase maps, amplitude spectra and Poincare maps, which were used to analyze and discuss the planar/non-planar multimodal coupled response of suspender. In addition, the non-planar modal coupled response of cross-flow and in-line directions was been discussed, too.

3. The influence of the non-planar modal interaction on the system response

The response diagram in Fig. 6 only considers the first-order mode of suspender and the self-excitation effect of aerodynamic load on the suspender, without taking into account the coupling effect of cross-flow and in-line directions. The analysis does not consider the structural nonlinearity caused by dynamic tension. The initial displacement/velocity conditions ($q_1 = 1\text{mm}; \dot{q}_1 = 0.1\text{mm/s}$) in both cross-flow and in-line directions are consistent.

Fig. 6 presents the time-history displacement diagram, phase diagram, amplitude spectrum diagram and the Poincare map of self-excitation process in the cross-flow and in-line directions. In Fig. (a), the cross-flow response and the in-line response is self-excited vibration under the action of wind load, the response amplitudes gradually reach steady-state equilibrium under the excitation of wind load and initial displacement/velocity conditions. In Fig. (b), the phase diagram was obtained by taking the response amplitude of a stable period and the corresponding response velocity. The phase diagram of the suspender response reaching the steady state is

Table 2
The physical parameters of suspender.

Parameter	Symbol	Units	Value
tension	H	N	2 636 000
length	l	m	114.78
elastic modulus	E	N/mm ²	83,177.6
diameter	D	m	0.142
wind velocity	U	m/s	3.000
mass per unit length	m	Kg/m	61.237
traverse area	S	mm ²	15,836.77
air mass density	ρ	Kg/m ³	1.2929
cross-flow dumping	μ_l	/	0.0014
in-line dumping	μ_d	/	0.0014

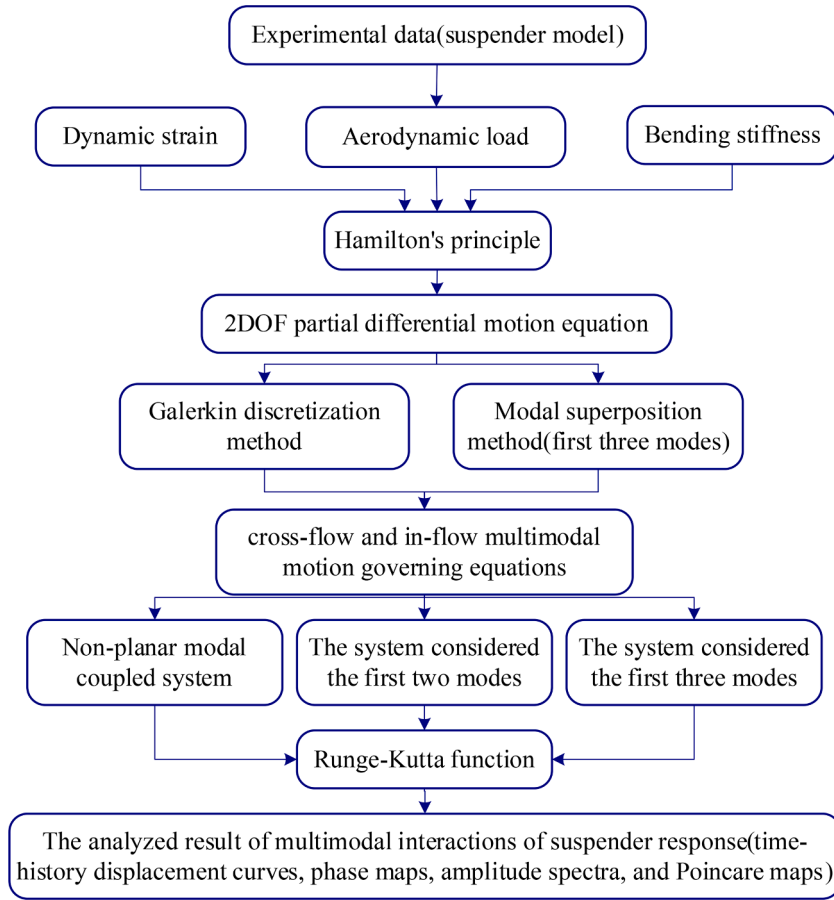


Fig. 5. The technical roadmap for suspender multimodal coupling research.

simple harmonic vibration, that is, the response amplitude and response velocity change regularly in the relationship of sine and cosine functions, which form a stable limit cycle. In Fig. (c), since there is only self-excited vibration, the amplitude spectrum only reflects the response amplitude corresponding to the natural frequency of the system. In Fig. (d), the response amplitude and response velocity were taken as a section (phase space); collected coordinates by taking a response amplitude every other natural frequency period and the response velocity corresponding to the response amplitude. Fig. (d) was obtained by gathering all the coordinates in 0–800 s on the section composed of the response amplitude and response velocity. When the system is stable, the points(response amplitude, response velocity) always fall on the limit cycle of steady-state phase diagram.

Fig. 7 is in agreement with Fig. 6, depicting time history displacement diagrams, phase diagrams, amplitude spectra and the Poincare maps that reflect the stabilization process in both cross-flow and in-line directions. The response diagram in Fig. 7 considers not only the aerodynamic load on the suspender but also the dynamic strain and bending stiffness of suspender. The suspender structural parameters are similar in the cross-flow and in-line directions, resulting in the same natural frequencies. The lock-in phenomenon occurs when the vortex shedding frequency is close to the structure frequency under the effect of wind load. The system response gradually reaches a stable state from the initial state under the joint effect of aerodynamic load and structural damping. In Fig. 7(a), the response amplitude of the cross-flow and in-line directions is a self-excited process under the effect of wind load. However, the response amplitude of Fig. 7(a) is 7 mm in both directions, which is lower than the response amplitude of Fig. 6(a) (the response amplitude is 9 mm and 4 mm in the cross-flow and in-line directions, respectively). This indicates that system energy in the cross-flow direction is transferred to the in-line direction under the effect of modal coupling effect caused by dynamic tension. In Fig. 7(b), the phase diagram was obtained by taking the response amplitude and response velocity of a period after stabilization. Comparing Fig. 6(b) with Fig. 7(b), it is worth pointing out that the phase diagram changes from a closed limit cycle to the period-seven periodic motion. After the response amplitude has stabilized, the time history displacement curve of the suspender should be taken ten seconds later and Fourier transform should be performed to obtain the amplitude spectrum diagram, as shown in Fig. 7(c). The Fig. 7(c) shows two harmonic double peaks in both the cross-flow and in-line responses. Compared to Fig. 6(c) and 7(c) presents a double frequency response that considers non-planar modal coupling effect. The amplitude spectrum shows that the response amplitude and response frequency of the cross-flow and in-line directions are very close. Through the analysis of Figs. 6(a) and 7(a), the system energy in the cross-flow direction is transferred to the in-line direction under the modal coupling effect caused by dynamic tension. The non-planar

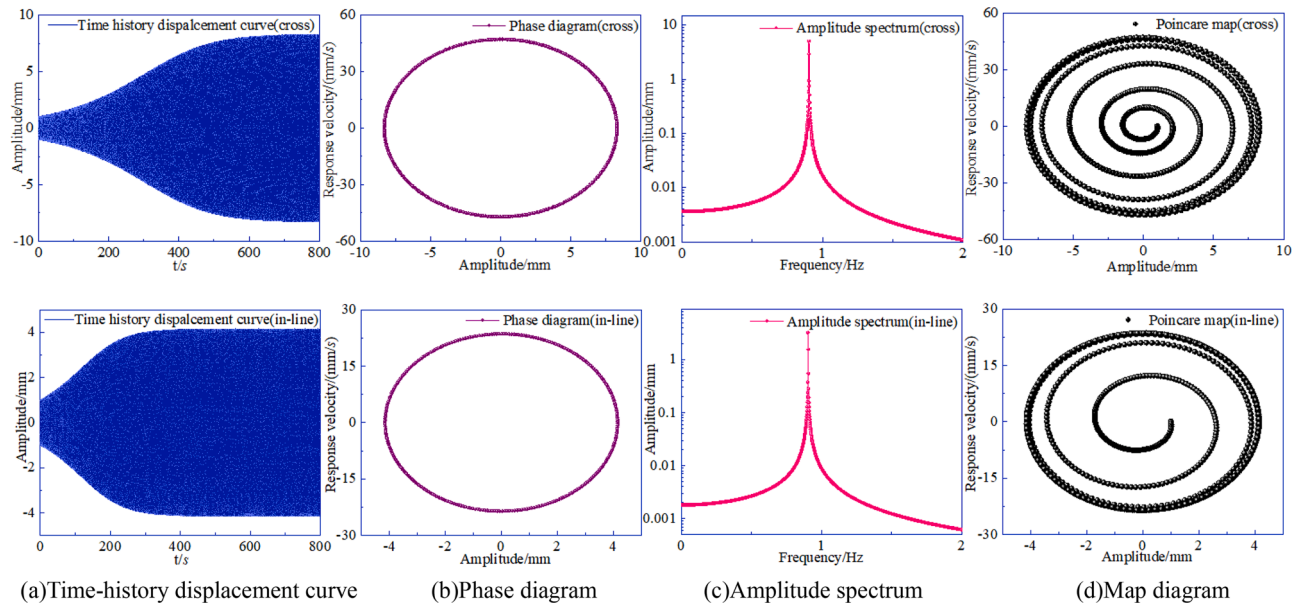


Fig. 6. Response diagram without considering non-planar modal coupling effect. (a)Time-history displacement curve (b)Phase diagram (c)Amplitude spectrum (d)Map diagram.

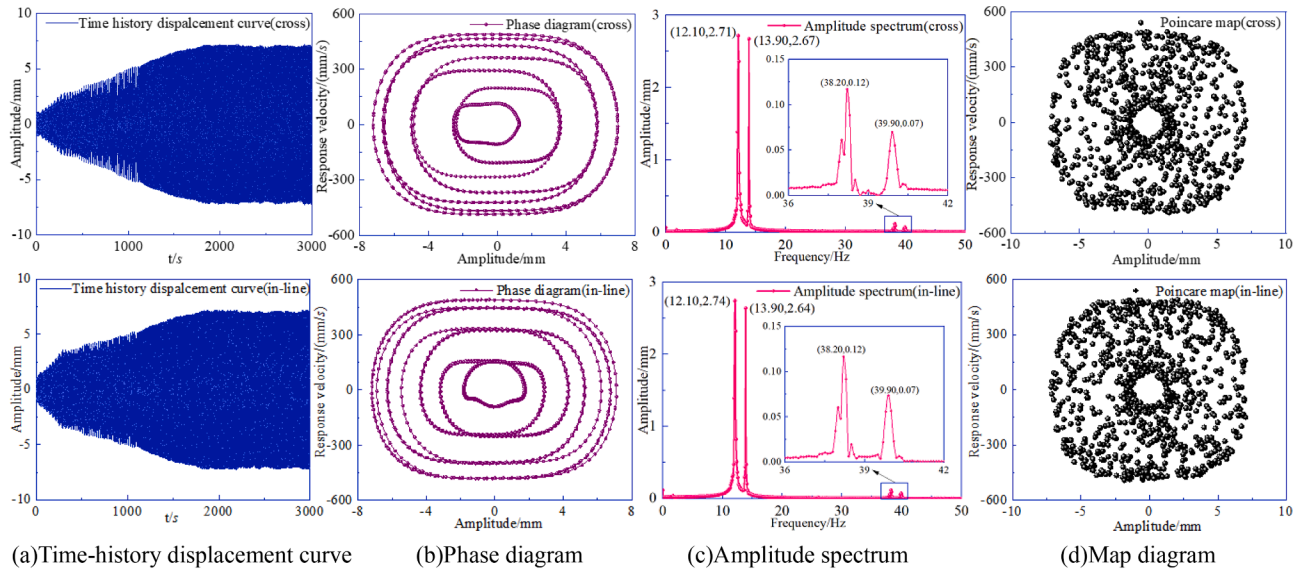


Fig. 7. Response diagram considering non-planar modal coupling effect. (a)Time-history displacement curve (b)Phase diagram (c)Amplitude spectrum (d)Map diagram.

modal 1:1 internal resonance is easily excited due to the frequency ratio of 1:1. Additionally, there is a sub-harmonic resonance effect of 3 times the dominant frequency in both directions. However, since the resonance effect of sub-harmonic resonance is much smaller than the amplitude of the dominant frequency, its influence on the system response can be ignored. Fig. 7(d) is a map of the cross-flow and in-line responses from 2000 to 3000 s, with mapping points filling a portion of the phase space and irregularly distributed within it. The map shows an unordered distribution of points, indicating that the response of the cross-flow and in-line directions appears the phenomenon of chaos.

From the analysis in Fig. 7, it can be seen that the dynamic tension leads to the non-planar modal 1:1 internal resonance. Under the coupling effect (1:1 internal resonance), the cross-flow and in-line responses amplitude is consistent due to energy transfer, the phase diagram shows multi-period vibration, accompanied by chaos. To compare and analyze the coupling effect, it ensured that the self-excitation effect of aerodynamic load in Fig. 7 is consistent with Fig. 6. Then, we obtain Fig. 8 by comparing the time history displacement curves of the cross-flow and in-line (0–500 s) in Fig. 7. In Fig. 8, the crests and troughs in the cross-flow and in-line directions correspond to each other, respectively, indicating a significant energy exchange phenomenon between the two directions. Due to the consistent parameters of the suspender in both directions, the ratio of natural frequencies in the cross-flow and in-line directions is 1:1, leading to an internal resonance between them under the non-planar coupling effect. This coupling effect is also known as non-planar 1:1 internal resonance effect.

As a result of the internal resonance effect, compared to Fig. 6, the time history displacement diagrams of the cross-flow and in-line responses are altered in Fig. 7. Furthermore, the nonlinear effects of these responses are significantly amplified in Fig. 7 compared to Fig. 6, their response amplitudes remain consistent. Experimental and numerical studies by reference [30] on wind-induced vibration of suspender also indicate that the amplitudes of the cross-flow and in-line responses are similar, confirming this theoretical model results. In addition, Fig. 7 shows harmonic interference near the primary resonance and the response amplitudes corresponding to the primary frequency and harmonic interference are very close. This observation indicates that the non-planar 1:1 internal resonance effect yields harmonic interference near the primary resonance in the cross-flow and in-line responses of suspender. Moreover, Figs. 7 and 8 reveal that the response amplitude of the suspender exhibits VIV-galloping phenomenon [31], which refers to that the suspender reaches the stable state after a self-excited process from an initial disturbance, where the response amplitude displays vortex-induced vibration phenomenon during this process.

Fig. 9 illustrates the response of suspender with a smaller elastic modulus. Subsequently, Fig. 9(a) presents the time history displacement curves in both the cross-flow and in-line directions. It is apparent that the response amplitude undergoes sudden changes at different time periods. Initially, the response amplitude undergoes self-excited vibration under the effect of vortex-induced force. Then, the waveform of the response amplitude showed vortex-induced vibration characteristics. The equilibrium position of the response amplitude gradually moves in the negative direction and the response amplitude undergoes several sudden changes resulting in crises and transient chaos [28]. In order to analyze the response characteristics of the response amplitude in Fig. (a), Fig. (b) takes time history displacement curves, phase diagrams and Poincare maps for different time periods. Given the very similar vibration characteristics of the cross-flow and in-line response amplitudes, only the response curves of different time periods in the cross-flow direction were analyzed. In Fig. (b), Part A displays the waveform of the time history displacement curves that exhibit vortex-induced characteristics, while the phase diagrams of Part B is an irregular limit cycle. Part C shows that the response amplitude forks and Part D indicates that the response amplitude has a period doubling bifurcation, changing from a limit cycle vibration with one period to a quasi-periodic vibration with two periods. After another sudden change, the response amplitude changes from a quasi-periodic vibration with two periods to periodic vibration with two periods in Part E. The Poincare map of the response amplitude in Part F is displayed as a scatter plot, Part G shows the response amplitude's irregular attenuation and Part H presents a point in Poincare map. It can be analyzed that under the effects of vortex-induced force, cubic nonlinearity and non-planar 1:1 internal resonance, the response amplitude first exhibits self-excitation phenomenon followed by vortex-induced vibration. Then, after a period doubling bifurcation, the response amplitude changes from a periodic vibration with one period to a period doubling vibration with two periods. After another sudden change, the response amplitude changes from a quasi-periodic vibration with two periods to a double periodic vibration with two periods. Then, the response amplitude rapidly decreases, and chaotic phenomena appear in the response amplitude. Finally, the response amplitude tends to a stable value after a transient chaos in response amplitude. It should be noted that there is still a strong coupling effect between the cross-flow and in-line responses while chaotic vibration disappears.

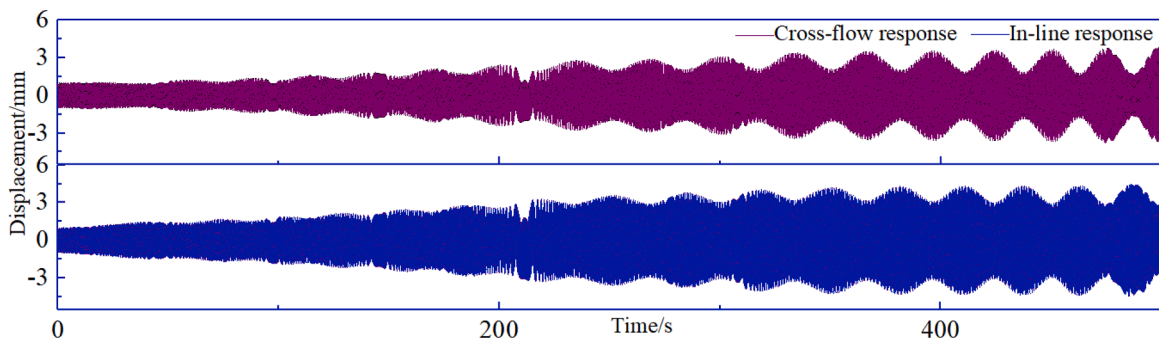
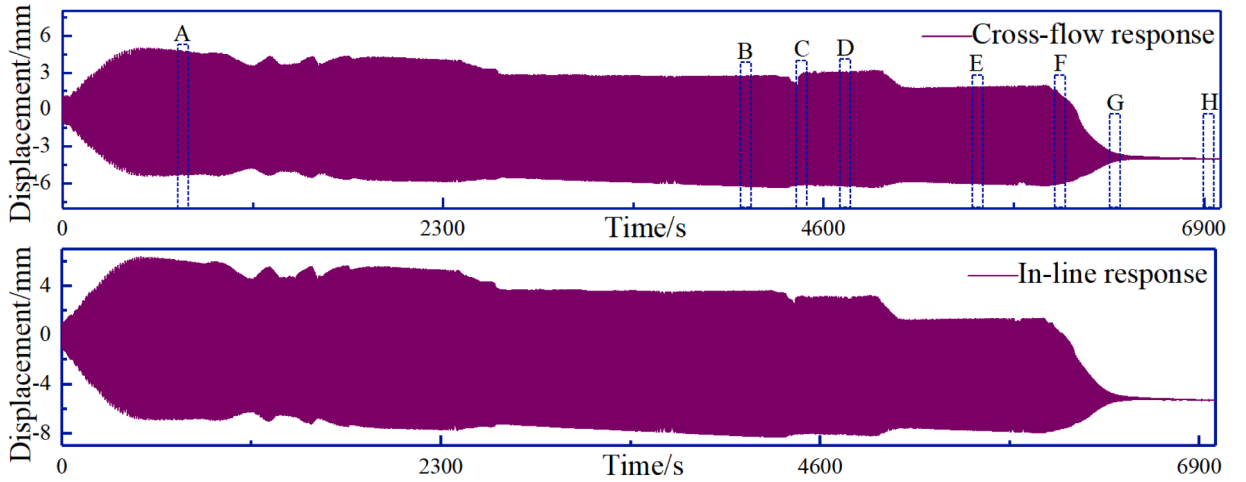
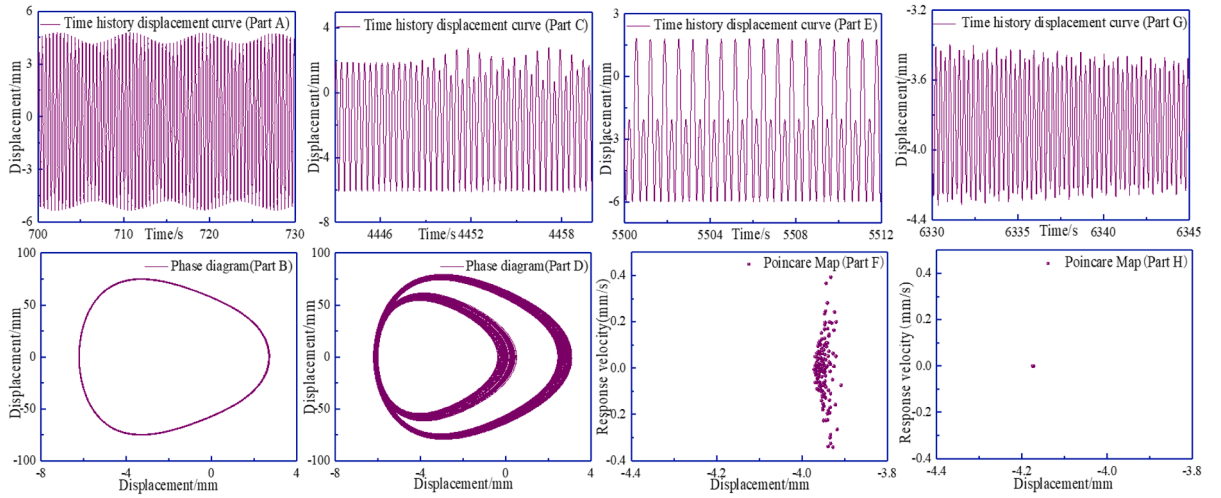


Fig. 8. Time history displacement curve.



(a) Time history displacement curve



(b) Response diagram

Fig. 9. Response diagram with small elastic modulus ($E = 1\,960$ Mpa). (a) Time history displacement curve (b) Response diagram.

Figs. 7 and 9 represent the response diagrams of suspender with different elastic modulus, namely $1\,960$ Mpa and $83\,177.6$ Mpa, respectively. When the elastic modulus is large, the nonlinear stiffness coupling effect of the suspender is greater, indicating that the enhancement of the nonlinear stiffness coupling effect leads to a significant enhancement of the internal resonance effect of suspender response. It also leads to an increase in response amplitude and an enhancement of the nonlinearity of suspender response. However, crises and transient chaos in Fig. 9 at different time periods only occur when the nonlinear stiffness coupling effect and the vortex-induced force effect are small.

4. The influence of the multi-modal interactions of suspenders on the dynamic response

To analyze the multi-modal coupling effect of suspender, we adopted the different elastic modulus and considered the coupling effect of the first two and first three modes. Figs. 10–12 show the time history displacement response of the first, first two and first three modes in the cross-flow and in-line directions, respectively. In Figs. 10–12, the elastic modulus in Fig. (a) is $1\,960$ Mpa and the elastic modulus in Fig. (b) is $83\,177.6$ Mpa, Figs. (a-b) show that the frequency ratio of the first-order mode in the cross-flow and in-line directions is 1:1, as is the case for the second-order mode and third-order mode; the ratio of the first-order mode frequency to the second-order mode frequency in the cross-flow direction in Figs. (a-b) is 1:2 and the ratio of the first-order mode frequency to the third-order mode frequency is 1:3, as is the case for the in-line direction.

In Fig. 10, under the modal coupling effect, the cross-flow and in-line responses in Figs. (a-b) were similar and there was a beat phenomenon in both cases. With time growing, the system response gradually became stable and the beat frequency gradually

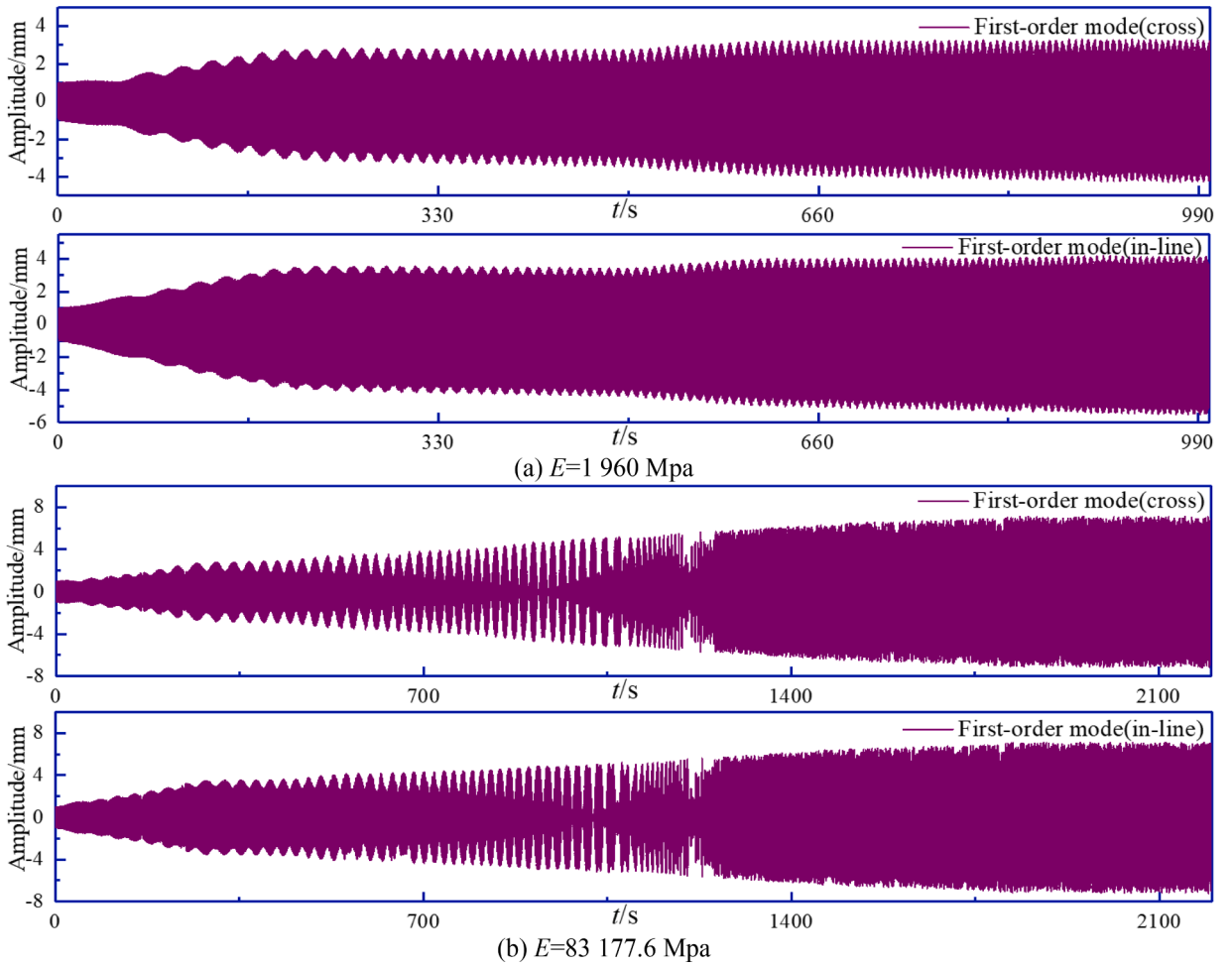


Fig. 10. The time-history displacement curve of first-order mode in cross-flow and in-line directions.

increased. In Figs. 10(a-b), there was a significant energy exchange phenomenon in the cross-flow and in-line responses. The response frequency in the cross-flow and in-line directions was 1:1, indicating that there was non-planar modal 1:1 internal resonance in Fig. 10 (a-b). It can be seen from Fig. 10 that the elastic modulus of the suspender in Fig. 10(a) was smaller than that in Fig. 10(b), indicating that the nonlinear effect of Fig. 10(b) was stronger than that in Fig. 10(a). The greater the elastic modulus of suspender, the more likely it yields nonlinear vibration. The change of the elastic modulus will lead to the change of the response results of the suspender. Although the response amplitude corresponding to Fig. 10(a) is smaller than that corresponding to Fig. 10(b), under the non-planar modal 1:1 internal resonance, the response amplitude of suspender in the cross-flow direction and in-line direction still remains the same.

In Fig. 11, the coupling effect of the first two modes is considered, resulting in the non-planar 1:1 internal resonance of the cross-flow and in-line second-order modal response. The time history displacement response of the cross-flow and in-line first-order modes remains consistent. It is noteworthy that the second-order modal response in Fig. 11(a) gradually decays to 0 during the process of the system response amplitude reaching stable, while the first-order modal response becomes predominant. The first-order modal response amplitude in Fig. 11(a) is greater than that in Fig. 11(b), while the second-order modal response in Fig. 11(b) is greater than the second-order modal response amplitude in Fig. 11(a). The modal coupling effect in Fig. 11(b) is greater than that in Fig. 11(a), indicating that the system energy of the first-order modal response in Fig. 11(b) is transferred to the second-order modal response. For the first-order modal response and second-order modal response with the same response amplitude, the frequency of the second-order modal response is greater than that of the first-order modal response, indicating that the second-order modal response plays a dominant role in system response. Therefore, for the suspender, the greater the elastic modulus (the stronger the modal coupling effect), the stronger the energy transfer between the low-order and high-order modes.

In Fig. 12, the coupling effect of the first three modes was been considered. The stable state of the cross-flow first-order modal response and the in-line first-order modal response is consistent, as is the case for the third-order modal response. In Fig. 12(a), the third-order modal response in Fig. 12(a) shows the form of irregular beating; the third-order modal response amplitude gradually attenuates to 0 during the process of system vibration, as well as the second-order modal response; only the first-order modal response

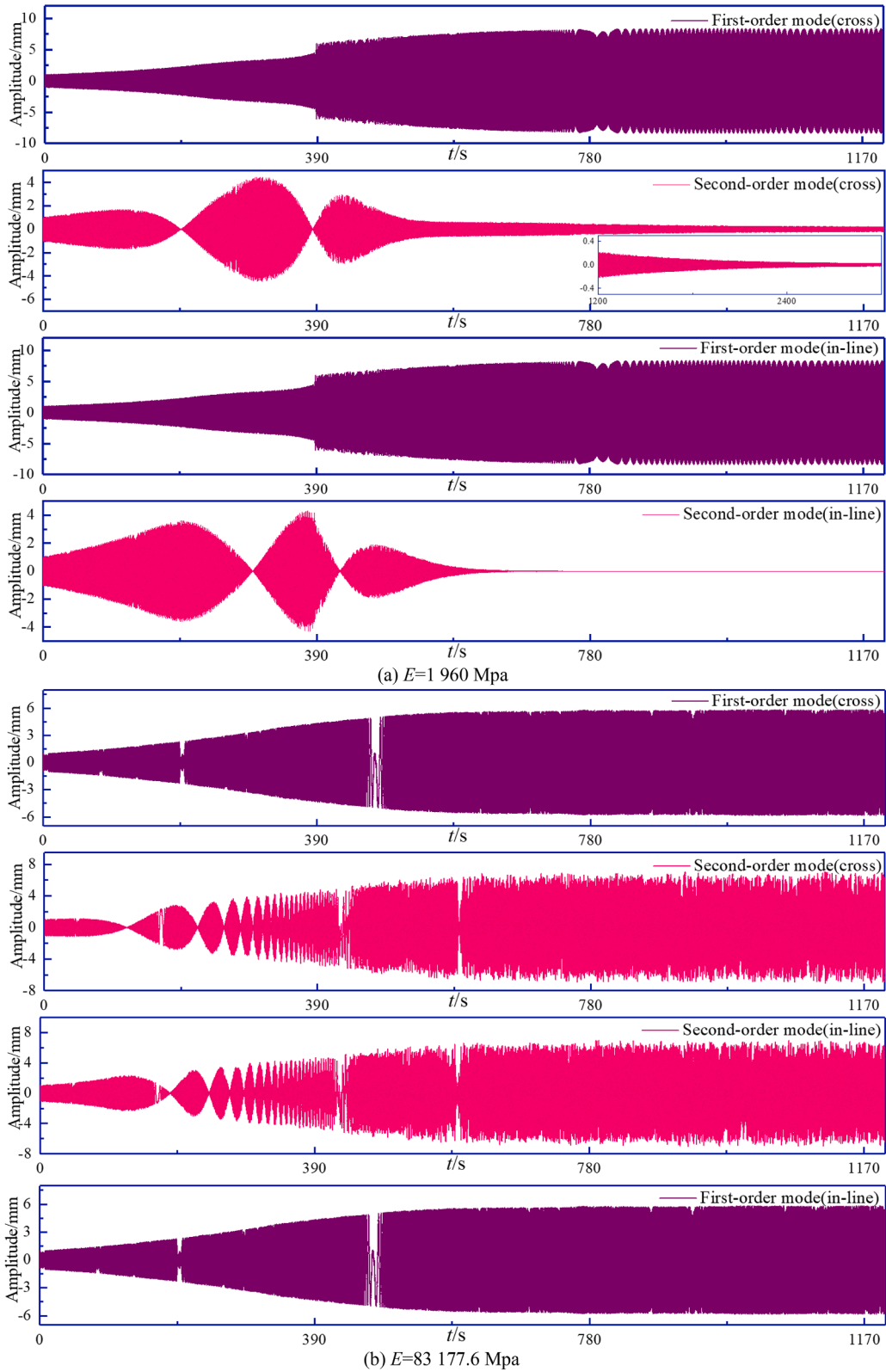


Fig. 11. The time-history displacement curve of different-order mode in cross-flow and in-line direction considering the coupling effect of the first two modes.

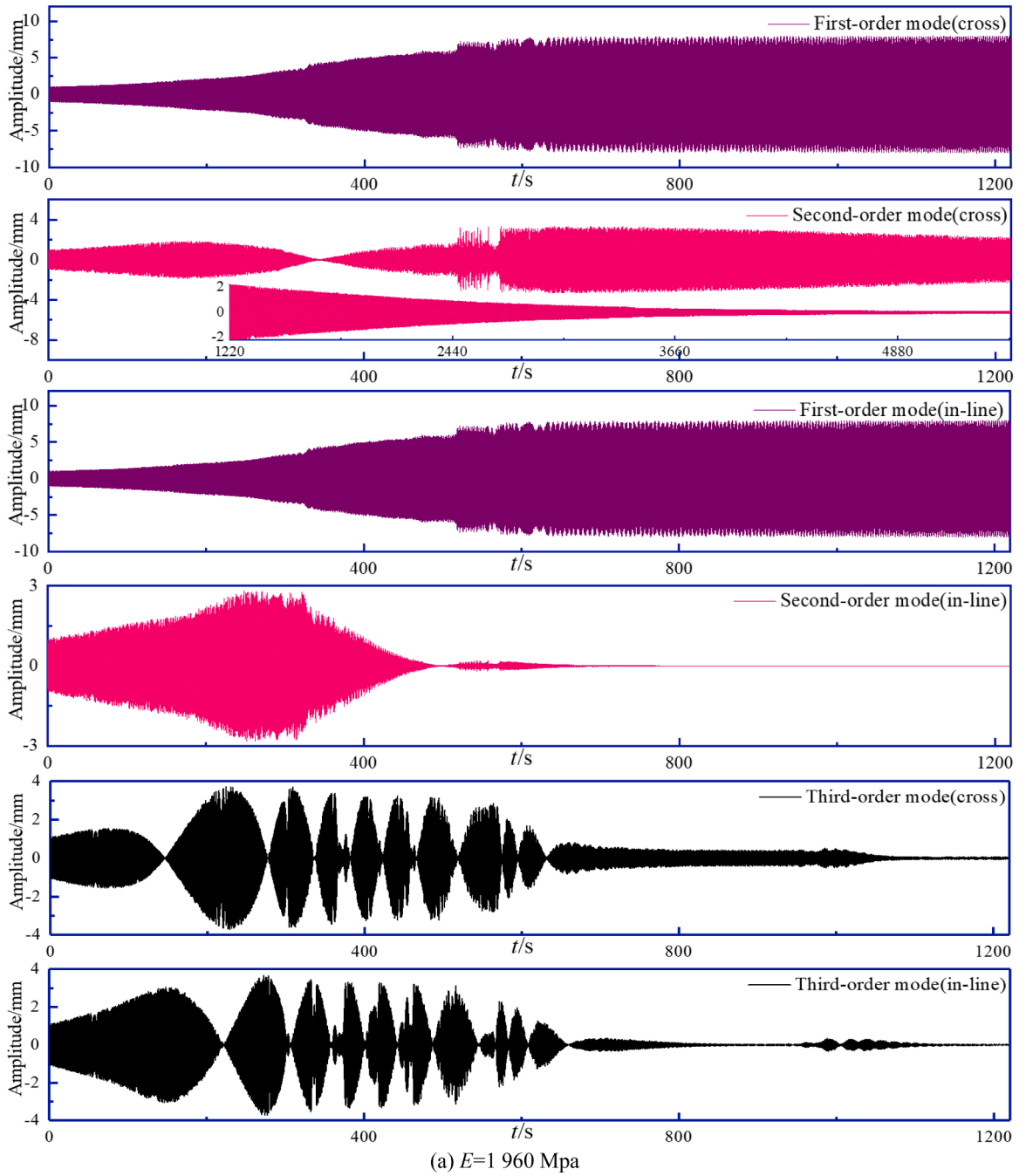


Fig. 12. The time-history displacement curve of different-order mode in cross-flow and in-line direction considering the coupling effect of the first three modes.

gradually reaches a stable state under the initial displacement/velocity conditions. This means that the system energy in the cross-flow and in-line direction is transferred from the second-order modal response and the third-order modal response to the first-order modal response, which is the planar multimodal 2:1 internal resonance effect and planar multimodal 3:1 internal resonance effect. In Fig. 12 (a), the crests and troughs of the third-order cross-flow and in-line responses correspond to each other, indicating energy exchange in the cross-flow and in-line third-order modal responses, that is the non-planar third-order modal 1:1 internal resonance. Fig. 12(b) illustrates that the wave crest of the first-order modal response corresponds to the wave trough of the third-order modal response, and vice versa. This indicates that there is energy exchange between the first-order and third-order modal responses, resulting in the planar multimodal 1:3 internal resonance in both the cross-flow and in-line directions. Additionally, the third-order modal response exhibits

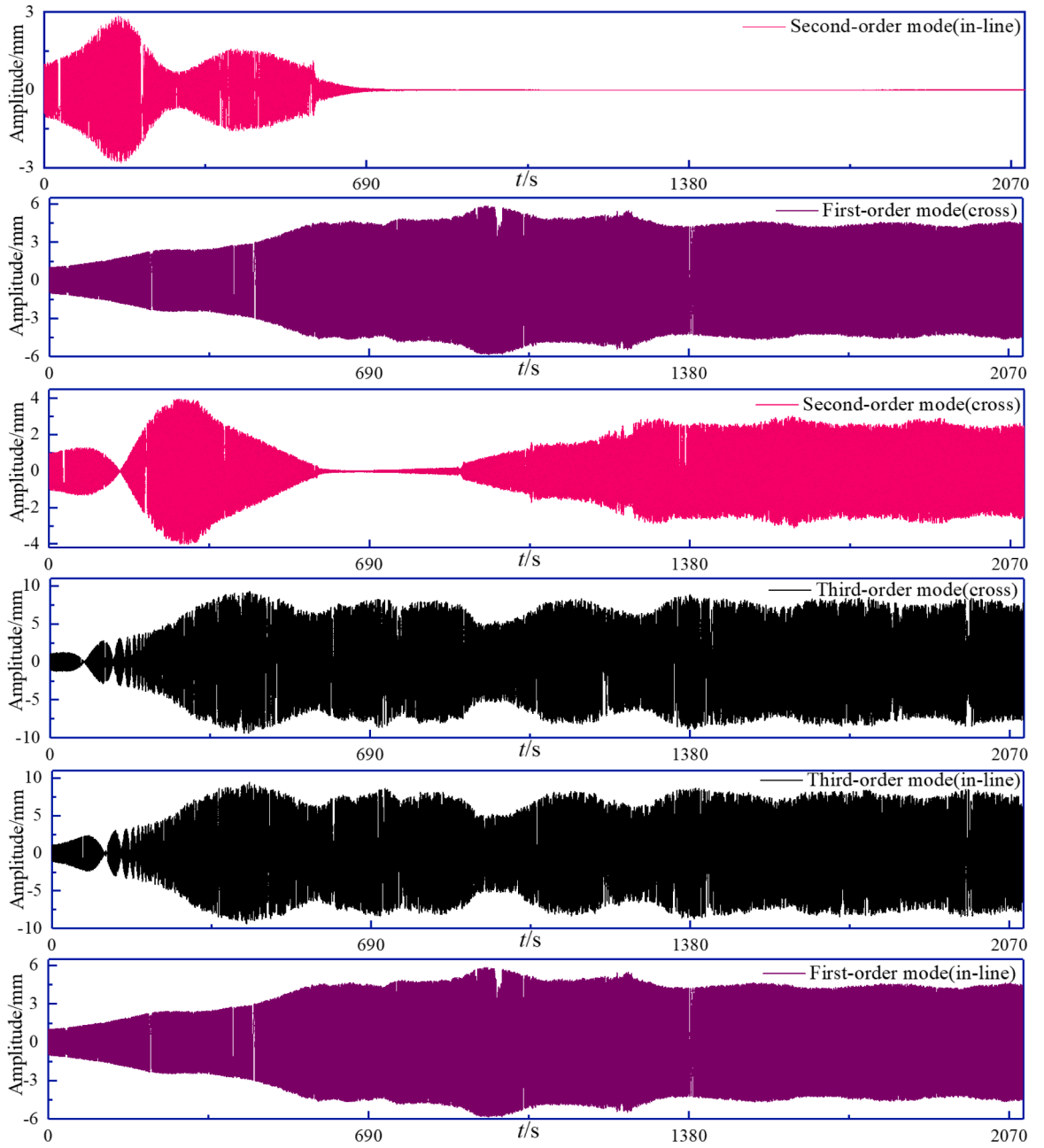
(b) $E=83\ 177.6\ \text{Mpa}$

Fig. 12. (continued).

energy exchange in the cross-flow and in-line modal response under initial displacement/velocity conditions in the initial stage, resulting in the non-planar third-order modal 1:1 internal resonance. As the system response gradually reaches a stable state, the energy conversion phenomenon of the third-order modal response and the first-order modal response becomes more apparent than the energy conversion phenomenon of the cross-flow third-order modal response and in-line third-order modal response in the initial stage. Furthermore, in the process of achieving stability of the high-order modal response, the system plays a dominant role in the non-planar third-order modal 1:1 internal resonance at the initial stage. In the stable stage, the planar multimodal 1:3 internal resonance of the system dominates.

In Fig. 12, the system energy is transferred not only between low-order and high-order modal responses (planar multimodal 1:3 internal resonance), but also between the cross-flow and in-line direction with the same-order modal response (non-planar third-order modal 1:1 internal resonance). Additionally, the enhanced coupling effect makes the high-order modal response dominate the system

response under the combined action of non-planar third-order modal 1:1 internal resonance and planar multimodal 1:3 internal resonance. In Fig. 12(b), after 800 s, during the self-excitation process of cross-flow second-order modal response, the cross-flow response amplitudes of both the first-order and third-order modal responses decreased (it can also be seen from the comparison diagram in Fig. 14(b)). There is an energy exchange between the cross-flow second-order modal response and not only the cross-flow first-order modal response, but also the cross-flow third-order modal response. This indicates the presence of non-planar/planar multimodal 1:2 internal resonance effect and planar multimodal 3:2 internal resonance effects in the second-order modal response.

For the suspender with small elastic modulus, the vibration characteristics of the first-order modal response in the cross-flow and in-line directions are very similar in Figs. 11(a) and 12(a). However, the first-order modal response in Fig. 12(a) reaches a stable state in a shorter time. This is why the system energy in Fig. 12(a) is transferred from the second-order modal response and the third-order

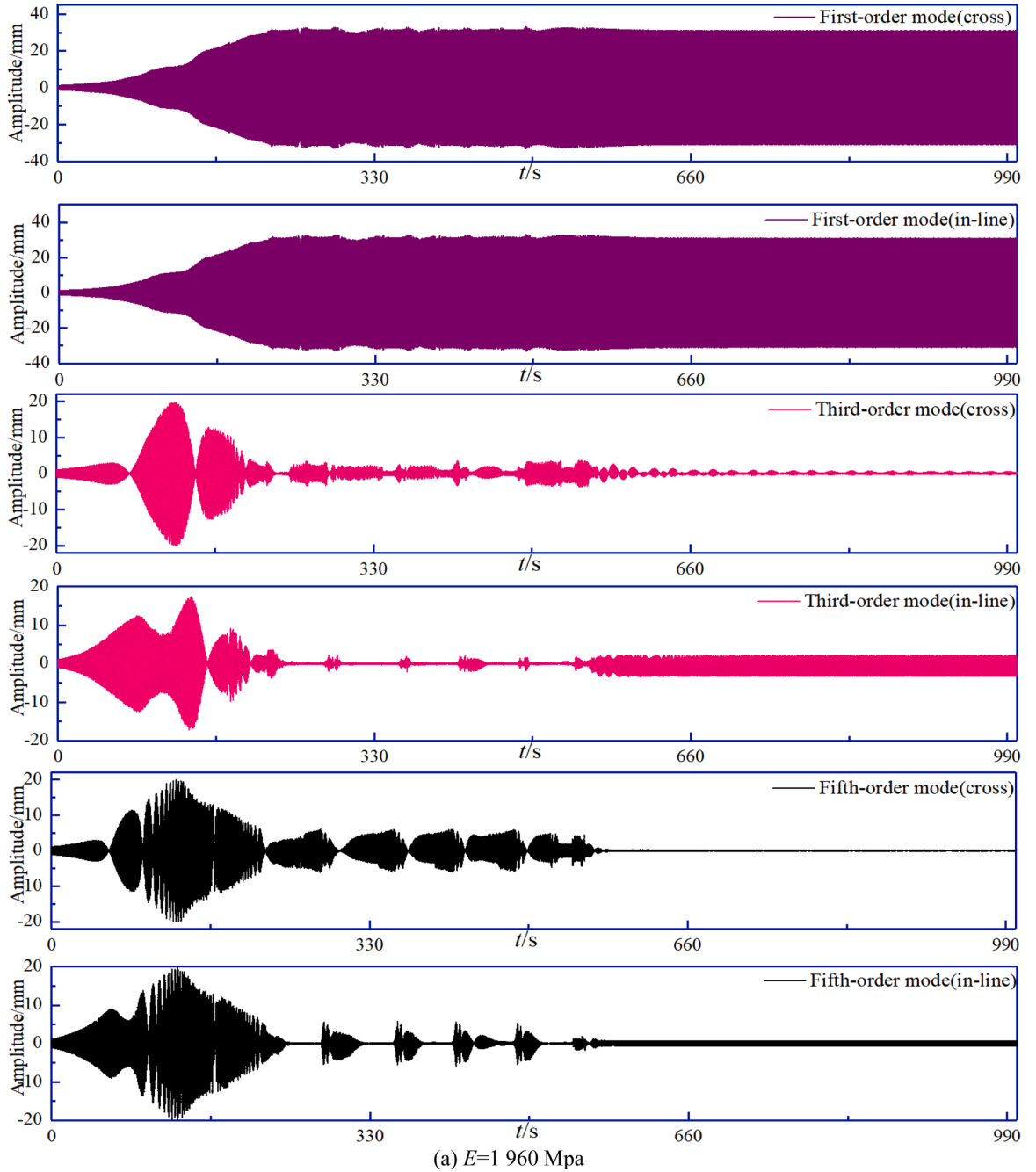


Fig. 13. The time-history displacement curve of different-order mode in cross-flow and in-line direction considering the coupling effect of the first-order, third-order and fifth-order modes.

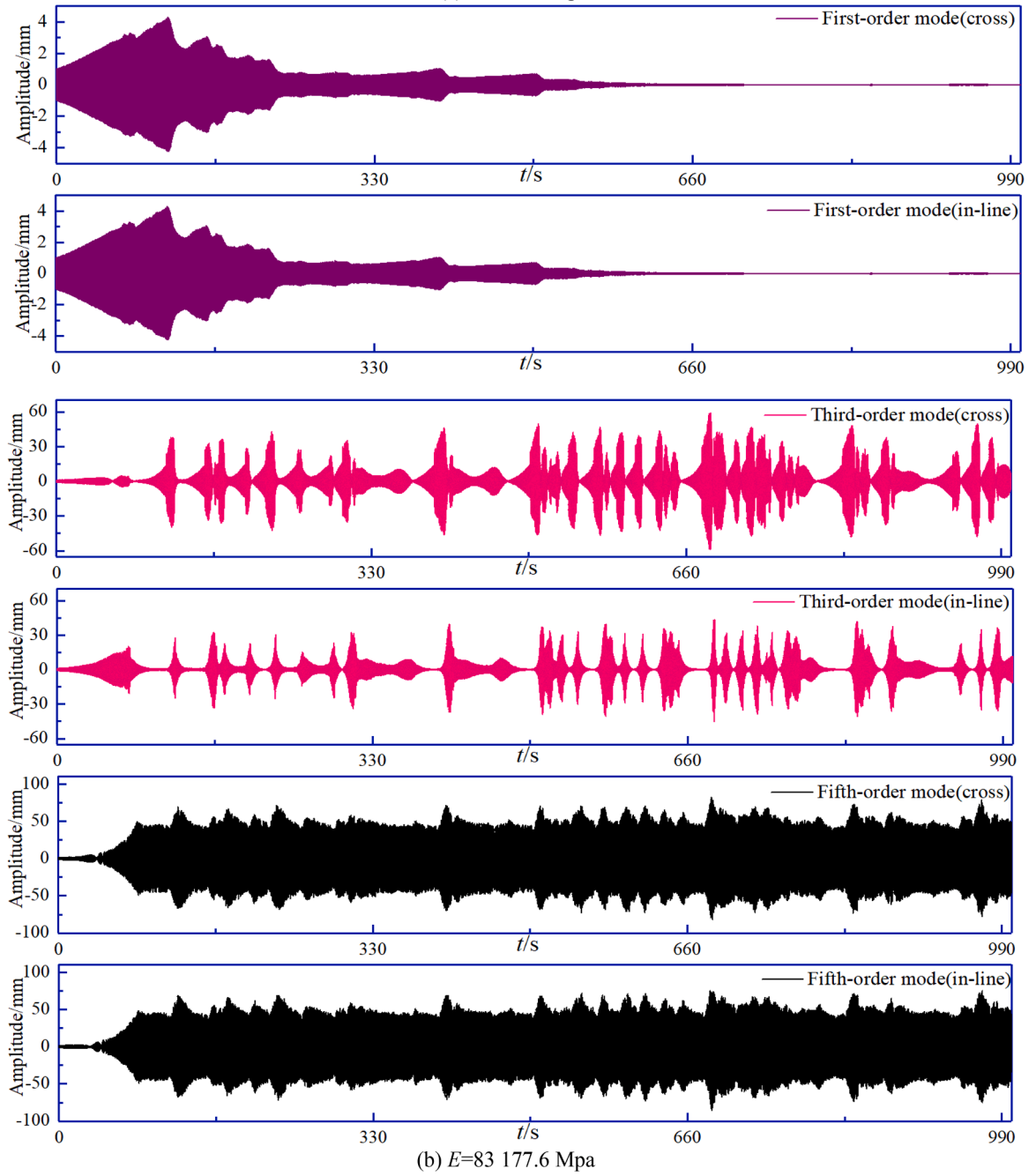


Fig. 13. (continued).

modal response to the first-order modal response, resulting in a more obvious internal resonance effect. Overall, the enhanced coupling effect is conducive to the energy transfer between low-order modes and high-order modes. With an increase in elastic modulus, the system energy is transferred from the first-order modal response to the third-order modal response. The high-order modal response can produce non-planar modal 1:1 resonance response faster under initial displacement/velocity conditions, resulting in the response amplitude reaching stable state in a shorter time.

To validate the aforementioned findings, the first-order mode, third-order mode and fifth-order mode of the cross-flow and in-line directions were considered, respectively. It's assumed that the other modes decay rapidly when only the interaction of these three modes was considered. The governing equation was obtained by introducing the first-order, third-order and fifth-order mode in Eq. (11b) into Eq. (12). Fig. 13 was obtained by solving the governing equation numerically. The elastic modulus in Fig. 13 are consistent with those in Figs. 10–12.

Fig. 13 reveals that the first-order modal response dominates the system response when the elastic modulus is small, whereas the fifth-order modal response dominates the system response when the elastic modulus is large. Notably, the non-planar multimodal 1:1 internal resonance significantly contributes to the system response. When the elastic modulus is small, the energy of the system response transfer from the high-order modal response to the low-order modal response, resulting in planar multimodal 3:1 and 5:1 internal resonance. This conclusion is consistent with the aforementioned findings. It is noteworthy that the system response exhibits richer nonlinear behavior as the elastic modulus increases. Specifically, when the elastic modulus is large, the contribution of the first-order modal response to the system is negligible compared to the contributions of the third-order and fifth-order modal responses. As the elastic modulus increases, the coupling effect of the suspender in both cross-flow and in-line directions is enhanced. A smaller coupling effect facilitates the control of system response by the cross-flow and in-line low-order modal responses of the suspender. Conversely, a larger coupling effect tends to excite high-order modal responses in the cross-flow and in-line directions of the suspender. Therefore, the strength of coupling effect should be taken into consideration when designing and controlling the system response.

The time history displacement curves presented in Fig. 14(a) and (b) can be obtained by introducing the cross-flow and in-line modal responses of Figs. 11(b) and Fig. 12(b) into Eq. (10), respectively. The amplitude spectrum of the cross-flow and in-line can be obtained by using the fast Fourier transform on the time-history displacement curve. Figs. 14(a) and (b) are the suspender response considering the second-order modal coupling effect and the third-order modal coupling effect, respectively.

In contrast, Fig. 7 only considers the coupling effect of non-planar first-order mode, the chaotic phenomena appear in their cross-flow and in-line responses. In Fig. 14, not only non-planar multimodal coupling effect but also the planar multimodal coupling effect are considered. It is evident that in Fig. 14, both the suspender responses considering the coupling effect of the first two modes and the coupling effect of the first three modes exhibit the phenomenon of chaos. The vibration characteristics of the cross-flow and in-line responses are highly similar, which is closely related to the consistency of the structural parameters of the suspender in the cross-flow and in-line directions. For the first-order mode, the frequency corresponding to the harmonic peak value of the amplitude spectrum considering the second-order modal coupling effect and the third-order modal coupling effect gradually increases, which is closely related to the coupling effect of the different modal responses.

Fig. 7 depicts a system comprising of a vortex-induced force, cubic nonlinear term, cubic nonlinear stiffness coupling term, disturbance excitation of near-primary resonance, with the most notable being non-planar modal 1:1 internal resonance. In Fig. 14(a), the system includes all items in Fig. 7, along with coupling terms of the first-order modal response and the second-order modal response in the cross-flow direction and the in-line direction, where the most prominent are non-planar modal 1:1 internal resonance and planar multimodal 1:2 internal resonance. Furthermore, Fig. 14(b) includes all items in Fig. 14(a), with the coupling terms of the cross-flow and in-line third-order modal response, first-order modal response and second-order modal response, among which the more obvious are non-planar modal 1:1 internal resonance, planar multimodal 1:2 internal resonance, planar multimodal 1:3 internal resonance.

The response diagram in Fig. 14(a) corresponds to the modal response diagrams in Fig. 11(b). The amplitude spectrum in Fig. 14(a) exhibits a harmonic response that is twice as large as the primary frequency. Similar to Fig. 7(c), the harmonic response in Fig. 14(a) also belongs to a harmonic disturbance. The harmonic response of the amplitude spectrum in Fig. 14(a) belongs to the 1:1 internal resonance in the non-planar second-order modes based on the analysis of Figs. 6(c), 7(c), and 11(b). There is also an energy exchange between the cross-flow and in-line second-order modal responses, confirming the existence of internal resonance between them.

The time history displacement curve in Fig. 14(b) is similar to VIV-galloping discussed in paper [24], but with stronger nonlinearity. In addition, the amplitude spectrum in Fig. 14(b) also shows similar characteristics to Fig. 12(b), the cross-flow direction has first-order, second-order and third-order modal responses; the in-line direction has only first-order and third-order modal responses. The first-order and third-order modal response amplitudes in the cross-flow direction are both smaller than those in the in-line direction, indicating that the part of system energy in the cross-flow response transfers from the first-order and third-order modal responses to the second-order modal responses. Thus, in a multi-degree of freedom system considering multimodal coupling effect, system energy can transfer not only from high-order mode to low-order mode, but also from low-order mode to high-order mode. The amplitude spectrum in Fig. 14(b) indicates that the third-order modal response dominates the system response, the cross-flow second-order modal response has a vital contribution to the cross-flow response compared with the in-line response.

Figs. 7 and 14(a), and (b) correspond to the first-order, second-order and third-order modal responses of suspender, respectively. The first-order modal response exhibits harmonic disturbances near the first-order primary resonance, the second-order modal response exhibits harmonic perturbations near the second-order resonance; the third-order modal response exhibits harmonic perturbations near the third-order resonance. Therefore, considering the first two modes and the first three modes of the suspender response, there is a second-order and third-order non-planar 1:1 internal resonance effect, respectively. That is, the 1:1 internal resonance effect is mainly reflected in the amplitude spectrum through harmonic disturbances near primary resonance; under strong coupling effect, the response of multimodal coupling systems is mainly reflected in the internal resonance of higher-order modal responses.

5. Conclusion

In this work, the vortex-induced parameters of suspender were obtained through an experimental model. By considering the vortex-induced force, dynamic tension and bending stiffness of suspender, the governing equation for the multimodal coupling effect of suspender is derived. Numerical solutions of the motion governing equation were utilized to obtain the non-planar modal coupled response of the suspender and the multi-modal coupled response of the suspender considered the first two and first three modal coupling effect. The results of this work were analyzed through time history displacement curves, phase diagrams, amplitude spectra,

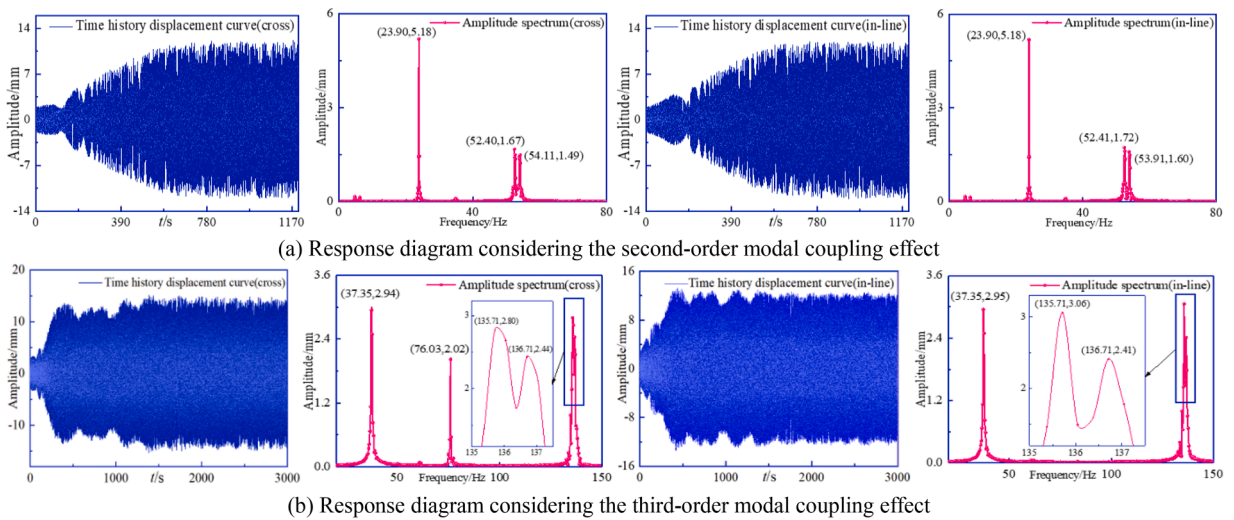


Fig. 14. Time history displacement diagram and amplitude spectrum diagram. (a) Response diagram considering the second-order modal coupling effect (b) Response diagram considering the third-order modal coupling effect.

and Poincare maps. The conclusions drawn from the analysis of the multimodal coupling response of the suspender were as follows:

- (1) Under the coupling effect caused by the dynamic tension of suspender, the cross-flow response and in-line response are easy to excite non-planar modal 1:1 internal resonance, the cross-flow response and in-line response generate chaotic vibration. The non-planar modal 1:1 internal resonance effect manifests itself in the harmonic interference of near primary resonance in the cross-flow and in-line responses of suspender. This interference leads to an increase in the response amplitude, causing the VIV-galloping phenomenon. Furthermore, the contribution of the nonlinear stiffness coupling effect of the in-line response to the system response weakens the limiting effect of the nonlinear aerodynamic stiffness term on the system response, resulting in a convergence between the values of the cross-flow and in-line response amplitudes. When the coupling effect of suspender nonlinear stiffness and vortex-induced force is small, the response amplitude in the cross-flow and in-line directions changes periodically. This includes the self-excited phenomenon, vortex-induced vibration phenomenon, an irregular limit cycle vibration form with a phase diagram of one period, period doubling bifurcation, quasi-periodic vibration with two periods, period doubling vibration with two periods, crises and transient chaos. For the multi-modal coupled suspender response, when the coupling effect is small, the low-order modal system response is dominant; as the coupling effect increases, the high-order modal response becomes predominant.
- (2) For the suspender, the greater the elastic modulus (the stronger the modal coupling effect), the stronger the internal resonance effect between the low-order modal response and the high-order modal response.
 - (a) The weak multi-modal interactions:

The suspender model takes into account the coupling effects of the first two modes and the first three modes, the non-planar modal 1:1 internal resonance effect makes the first-order modal response dominate the system response. As the system response becomes stable, the cross-flow and in-line second-order and third-order modal responses reach the response peak in the form of irregular beats, which decay to 0 later.

(b) The strong multi-modal interactions:

- (i) When the suspender model considers the coupling effect of the first two modes: the planar multimodal 1:2 internal resonance effect makes the system energy transfer from the first-order mode response to the second-order mode response. The non-planar second-order modal 1:1 internal resonance effect and the planar multimodal 1:2 internal resonance effect make the second-order modal response dominate the system response. The non-planar first-order modal 1:1 internal resonance makes the first-order modal response a prominent contribution to the system response.
- (ii) When the suspender model considers the coupling effect of the first three modes: the planar multimodal 1:3 internal resonance effect transfers the system energy from the first-order mode response to the third-order mode response. The non-planar third-order modal 1:1 internal resonance effect and the planar multimodal 1:3 internal resonance effect make the

third-order modal response become predominant in the system response. The non-planar first-order modal 1:1 internal resonance effect makes the a significant contribution(first-order modal response) to the system response. In the cross-flow direction, the planar multimodal 1:2/3:2 internal resonance makes the system energy transfer from the first-order and three-order modal response to the second-order modal response and the second-order modal response make a contribution that cannot be ignored in the system response.

Declaration of Competing Interest

The authors declare that there are no conflicts of interest regarding the publication of this paper.

Data availability

Data will be made available on request.

Acknowledgments

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Appendix. A

Initial amplitude : a_0 ; Disturbance amplitude : B ; Natural frequency : ω_{d1} ; Disturbance frequency : ω_{d2} ;

Oscillator model :

$$\ddot{q} + \omega_{d1}^2 q + a\dot{q} + b\dot{q}q^2 + p \sin(\omega_{d2}t + \theta_1) = 0;$$

Analytical solution of oscillator model :

$$q = \Lambda \cdot \cos\psi + 2B \cdot \cos(\lambda\psi) + \varepsilon \left(\frac{1}{(8\omega_{d1}\lambda(9\lambda^6 - 91\lambda^4 + 91\lambda^2 - 9))} \cdot (-36aB^2\Lambda(\lambda^4 - 82\lambda^2/9 + 1)((\lambda^2 + \lambda/2 - 1/2) \cdot \sin((2\lambda - 1)\psi) + (\lambda^2 - \lambda/2 + 1/2) \cdot \sin((2\lambda + 1)\psi)) + 36\lambda b(\lambda^2 - 1/9)(\lambda^2 + \lambda - 6)\Lambda B \cdot \sin((2 - \lambda)\psi) - 36\lambda b(\lambda^2 - 1/9)(\lambda^3 - 2\lambda^2 - 5\lambda + 6)\Lambda^2 B \cdot \sin((2 + \lambda)\psi) + \lambda^2 b(\lambda^2 - 9)(\lambda^2 - 1)B^3 \cdot \sin(3\lambda\psi) + 144B\lambda^2(\lambda^4 - 82\lambda^2/9 + 1)(b\Lambda^2/2 + a/2 + bB^2/2) \cdot \sin(\lambda\psi) - 9\lambda b(\lambda^2 - 1)(b\Lambda^2/2 + a/2 + bB^2/2)\Lambda^3 \cdot \sin(3\lambda\psi)/4 \right);$$

Amplitude expression :

$$\Lambda = \sqrt{1/[(1/a_0^2 + n_1/r_1)e^{-2r_1t} - n_1/r_1]}; B = p \cdot \sqrt{[2|\omega_{d2}^2 - \omega_{d1}^2|]};$$

$$\psi = \omega_{d1}t + \beta_1; \lambda = \omega_{d2}/\omega_{d1}; n_1 = -\varepsilon b/8; r_1 = -\varepsilon(a/2 + bB^2); \beta_1 = \beta_0; \varepsilon = 0.1;$$

Appendix. B

$$\begin{aligned} I_1 &= \frac{ES}{2mI_i}; \omega_1^2 = \frac{I_{\omega 1}}{mI_i}; c_{\xi i} = I_1 I_{\xi i} (\xi = 1, 2 \dots 12); c_{13i} = \frac{a^* + \mu_d}{m}; I_2 = \frac{b^*}{mI_i}; c_{\xi i} = I_2 I_{\xi i} (\xi = 14, 15 \dots 23); c_{24i} = \frac{I_{pi} P}{mI_i}; \\ I_3 &= \frac{ES}{2mI_r}; \omega_2^2 = \frac{I_{\omega 2}}{mI_r}; d_{\xi i} = I_3 R_{\xi} (\xi = 1, 2 \dots 12); d_{13i} = \frac{a_1^* + \mu_l}{m}; I_4 = \frac{b_1^*}{mI_r}; d_{\xi i} = I_4 R_{\xi} (\xi = 14, 15 \dots 23); d_{24i} = \frac{I_{pr} P_1}{mI_r}; \\ I_5 &= \frac{ES}{2mI_j}; \omega_3^2 = \frac{I_{\omega 3}}{mI_j}; c_{\xi j} = I_5 J_{\xi} (\xi = 1, 2 \dots 12); c_{13j} = \frac{a^* + \mu_d}{m}; I_6 = \frac{b^*}{mI_j}; c_{\xi j} = I_6 J_{\xi} (\xi = 14, 15 \dots 23); c_{24j} = \frac{I_{pj} P}{mI_j}; \\ I_7 &= \frac{ES}{2mI_s}; \omega_4^2 = \frac{I_{\omega 4}}{mI_s}; d_{\xi j} = I_7 S_{\xi} (\xi = 1, 2 \dots 12); d_{13j} = \frac{a_1^* + \mu_l}{m}; I_8 = \frac{b_1^*}{mI_s}; d_{\xi j} = I_8 S_{\xi} (\xi = 14, 15 \dots 23); d_{24j} = \frac{I_{ps} P_1}{mI_s}; \\ I_9 &= \frac{ES}{2mI_k}; \omega_5^2 = \frac{I_{\omega 5}}{mI_k}; c_{\xi k} = I_9 K_{\xi} (\xi = 1, 2 \dots 12); c_{13k} = \frac{a^* + \mu_d}{m}; I_{10} = \frac{b^*}{mI_k}; c_{\xi k} = I_{10} K_{\xi} (\xi = 14, 15 \dots 23); c_{24k} = \frac{I_{pk} P}{mI_k}; \\ I_{11} &= \frac{ES}{2mI_t}; \omega_6^2 = \frac{I_{\omega 6}}{mI_t}; d_{\xi t} = I_{11} T_{\xi} (\xi = 1, 2 \dots 12); d_{13k} = \frac{a_1^* + \mu_l}{m}; I_{12} = \frac{b_1^*}{mI_t}; d_{\xi k} = I_{12} T_{\xi} (\xi = 14, 15 \dots 23); d_{24k} = \frac{I_{pt} P_1}{mI_t}; \end{aligned}$$

$$\begin{aligned}
I_i &= \int_0^l f_1^2 dx; I_{\omega 1} = H \int_0^l (f_1')^2 dx + EI \int_0^l f_1^{(4)} f_1 dx; I_{1i} = \int_0^l (f_1')^2 dx \int_0^l (f_1')^2 dx; I_{2i} = \int_0^l (f_1')^2 dx \int_0^l (f_2')^2 dx; \\
I_{3i} &= \int_0^l (f_1')^2 dx \int_0^l (f_3')^2 dx; I_{4i} = \int_0^l (f_1')^2 dx \int_0^l f_1' f_2' dx; I_{5i} = \int_0^l (f_1')^2 dx \int_0^l f_1' f_3' dx; I_{6i} = \int_0^l (f_1')^2 dx \int_0^l f_2' f_3' dx; \\
I_{7i} &= \int_0^l (f_1')^2 dx \int_0^l (\theta_1')^2 dx; I_{8i} = \int_0^l (f_1')^2 dx \int_0^l (\theta_2')^2 dx; I_{9i} = \int_0^l (f_1')^2 dx \int_0^l (\theta_3')^2 dx; I_{10i} = \int_0^l (f_1')^2 dx \int_0^l \theta_1' \theta_2' dx; \\
I_{11i} &= \int_0^l (f_1')^2 dx \int_0^l \theta_1' \theta_3' dx; I_{12i} = \int_0^l (f_1')^2 dx \int_0^l \theta_2' \theta_3' dx; I_{14i} = \int_0^l f_1^4 dx; I_{15i} = \int_0^l f_1 f_2^3 dx; I_{16i} = \int_0^l f_1 f_3^3 dx; \\
I_{17i} &= \int_0^l f_1^3 f_2 dx; I_{18i} = \int_0^l f_1^3 f_3 dx; I_{19i} = \int_0^l f_1^2 f_2^2 dx; I_{20i} = \int_0^l f_1 f_2^2 f_3 dx; I_{21i} = \int_0^l f_1 f_2 f_3^2 dx; I_{22i} = \int_0^l f_1^2 f_2^2 dx; \\
I_{23i} &= \int_0^l f_1^2 f_2 f_3 dx; I_{pi} = \int_0^l f_1 f_0 dx;
\end{aligned}$$

$$\begin{aligned}
I_r &= \int_0^l \theta_1^2 dx; I_{\omega 2} = H \int_0^l (\theta_1')^2 dx + EI \int_0^l \theta_1^{(4)} \theta_1 dx; R_1 = \int_0^l (\theta_1')^2 dx \int_0^l (\theta_1')^2 dx; R_2 = \int_0^l (\theta_1')^2 dx \int_0^l (\theta_2')^2 dx; \\
R_3 &= \int_0^l (\theta_1')^2 dx \int_0^l (\theta_3')^2 dx; R_4 = \int_0^l (\theta_1')^2 dx \int_0^l \theta_1' \theta_2' dx; R_5 = \int_0^l (\theta_1')^2 dx \int_0^l \theta_1' \theta_3' dx; R_6 = \int_0^l (\theta_1')^2 dx \int_0^l \theta_2' \theta_3' dx; \\
R_7 &= \int_0^l (f_1')^2 dx \int_0^l (\theta_1')^2 dx; R_8 = \int_0^l (\theta_1')^2 dx \int_0^l (f_2')^2 dx; R_9 = \int_0^l (\theta_1')^2 dx \int_0^l (f_3')^2 dx; R_{10} = \int_0^l (\theta_1')^2 dx \int_0^l f_1' f_2' dx; \\
R_{11} &= \int_0^l (\theta_1')^2 dx \int_0^l f_1' f_3' dx; R_{12} = \int_0^l (\theta_1')^2 dx \int_0^l f_2' f_3' dx; R_{14} = \int_0^l \theta_1^4 dx; R_{15} = \int_0^l \theta_1 \theta_2^3 dx; R_{16} = \int_0^l \theta_1 \theta_3^3 dx; \\
R_{17} &= \int_0^l \theta_1^3 \theta_2 dx; R_{18} = \int_0^l \theta_1^3 \theta_3 dx; R_{19} = \int_0^l \theta_1^2 \theta_2^2 dx; R_{20} = \int_0^l \theta_1 \theta_2^2 \theta_3 dx; R_{21} = \int_0^l \theta_1 \theta_2 \theta_3^2 dx; R_{22} = \int_0^l \theta_1^2 \theta_3^2 dx; \\
R_{23} &= \int_0^l \theta_1^2 \theta_2 \theta_3 dx; I_{pr} = \int_0^l \theta_1 \theta_0 dx;
\end{aligned}$$

$$\begin{aligned}
I_j &= \int_0^l f_2^2 dx; I_{\omega 3} = H \int_0^l (f_2')^2 dx + EI \int_0^l f_2^{(4)} f_2 dx; J_1 = \int_0^l (f_2')^2 dx \int_0^l (f_2')^2 dx; J_2 = \int_0^l (f_1')^2 dx \int_0^l (f_2')^2 dx; \\
J_3 &= \int_0^l (f_2')^2 dx \int_0^l (f_3')^2 dx; J_4 = \int_0^l (f_2')^2 dx \int_0^l f_1' f_2' dx; J_5 = \int_0^l (f_2')^2 dx \int_0^l f_1' f_3' dx; J_6 = \int_0^l (f_2')^2 dx \int_0^l f_2' f_3' dx; \\
J_7 &= \int_0^l (f_2')^2 dx \int_0^l (\theta_1')^2 dx; J_8 = \int_0^l (f_2')^2 dx \int_0^l (\theta_2')^2 dx; J_9 = \int_0^l (f_2')^2 dx \int_0^l (\theta_3')^2 dx; J_{10} = \int_0^l (f_2')^2 dx \int_0^l \theta_1' \theta_2' dx; \\
J_{11} &= \int_0^l (f_2')^2 dx \int_0^l \theta_1' \theta_3' dx; J_{12} = \int_0^l (f_2')^2 dx \int_0^l \theta_2' \theta_3' dx; J_{14} = \int_0^l f_2 f_1^3 dx; J_{15} = \int_0^l f_2^4 dx; J_{16} = \int_0^l f_2 f_3^3 dx; \\
J_{17} &= \int_0^l f_1^2 f_2^2 dx; J_{18} = \int_0^l f_1 f_2^2 f_3 dx; J_{19} = \int_0^l f_1 f_2^2 dx; J_{20} = \int_0^l f_2^3 f_3 dx; J_{21} = \int_0^l f_2^2 f_3^2 dx; J_{22} = \int_0^l f_1 f_2 f_3^2 dx; \\
J_{23} &= \int_0^l f_1 f_2^2 f_3 dx; I_{pj} = \int_0^l f_2 f_0 dx;
\end{aligned}$$

$$\begin{aligned}
I_r &= \int_0^l \theta_2^2 dx; I_{\omega 4} = H \int_0^l (\theta_2')^2 dx + EI \int_0^l \theta_2^{(4)} \theta_2 dx; S_1 = \int_0^l (\theta_2')^2 dx \int_0^l (\theta_2')^2 dx; S_2 = \int_0^l (\theta_1')^2 dx \int_0^l (\theta_2')^2 dx; \\
S_3 &= \int_0^l (\theta_2')^2 dx \int_0^l (\theta_3')^2 dx; S_4 = \int_0^l (\theta_2')^2 dx \int_0^l \theta_1' \theta_2' dx; S_5 = \int_0^l (\theta_2')^2 dx \int_0^l \theta_1' \theta_3' dx; S_6 = \int_0^l (\theta_2')^2 dx \int_0^l \theta_2' \theta_3' dx; \\
S_7 &= \int_0^l (f_1')^2 dx \int_0^l (\theta_2')^2 dx; S_8 = \int_0^l (\theta_2')^2 dx \int_0^l (f_2')^2 dx; S_9 = \int_0^l (\theta_2')^2 dx \int_0^l (f_3')^2 dx; S_{10} = \int_0^l (\theta_2')^2 dx \int_0^l f_1' f_2' dx; \\
S_{11} &= \int_0^l (\theta_2')^2 dx \int_0^l f_1' f_3' dx; S_{12} = \int_0^l (\theta_2')^2 dx \int_0^l f_2' f_3' dx; S_{14} = \int_0^l \theta_1^3 \theta_2 dx; S_{15} = \int_0^l \theta_2^4 dx; S_{16} = \int_0^l \theta_2 \theta_3^3 dx; \\
S_{17} &= \int_0^l \theta_1^2 \theta_2^2 dx; S_{18} = \int_0^l \theta_1^2 \theta_2 \theta_3 dx; S_{19} = \int_0^l \theta_1 \theta_2^2 \theta_3 dx; S_{20} = \int_0^l \theta_2^3 \theta_3 dx; S_{21} = \int_0^l \theta_2^2 \theta_3^2 dx; S_{22} = \int_0^l \theta_1 \theta_2 \theta_3^2 dx; \\
S_{23} &= \int_0^l \theta_1 \theta_2^2 \theta_3 dx; I_{ps} = \int_0^l \theta_2 \theta_0 dx;
\end{aligned}$$

$$\begin{aligned}
I_k &= \int_0^l f_3^2 dx; I_{\omega 5} = H \int_0^l (f_3')^2 dx + EI \int_0^l f_3^{(4)} f_3 dx; K_1 = \int_0^l (f_3')^2 dx \int_0^l (f_3')^2 dx; K_2 = \int_0^l (f_3')^2 dx \int_0^l (f_2')^2 dx; \\
K_3 &= \int_0^l (f_1')^2 dx \int_0^l (f_3')^2 dx; K_4 = \int_0^l (f_3')^2 dx \int_0^l f_1' f_3' dx; K_5 = \int_0^l (f_3')^2 dx \int_0^l f_2' f_3' dx; K_6 = \int_0^l (f_3')^2 dx \int_0^l f_1' f_2' dx; \\
K_7 &= \int_0^l (f_3')^2 dx \int_0^l (\theta_1')^2 dx; K_8 = \int_0^l (f_3')^2 dx \int_0^l (\theta_2')^2 dx; K_9 = \int_0^l (f_3')^2 dx \int_0^l (\theta_3')^2 dx; K_{10} = \int_0^l (f_3')^2 dx \int_0^l \theta_1' \theta_2' dx; \\
K_{11} &= \int_0^l (f_3')^2 dx \int_0^l \theta_1' \theta_3' dx; K_{12} = \int_0^l (f_3')^2 dx \int_0^l \theta_2' \theta_3' dx; K_{14} = \int_0^l f_3 f_1^3 dx; K_{15} = \int_0^l f_3 f_2^3 dx; K_{16} = \int_0^l f_3^4 dx; \\
K_{17} &= \int_0^l f_1^2 f_2 f_3 dx; K_{18} = \int_0^l f_1^2 f_3^3 dx; K_{19} = \int_0^l f_1 f_2^2 f_3 dx; K_{20} = \int_0^l f_2^2 f_3^3 dx; K_{21} = \int_0^l f_2 f_3^3 dx; K_{22} = \int_0^l f_1 f_3^3 dx; \\
K_{23} &= \int_0^l f_1 f_2 f_3^2 dx; I_{pk} = \int_0^l f_3 f_0 dx; \\
I_l &= \int_0^l \theta_3^2 dx; I_{\omega 6} = H \int_0^l (\theta_3')^2 dx + EI \int_0^l \theta_3^{(4)} \theta_3 dx; T_1 = \int_0^l (\theta_3')^2 dx \int_0^l (\theta_3')^2 dx; T_2 = \int_0^l (\theta_1')^2 dx \int_0^l (\theta_3')^2 dx; \\
T_3 &= \int_0^l (\theta_2')^2 dx \int_0^l (\theta_3')^2 dx; T_4 = \int_0^l (\theta_3')^2 dx \int_0^l \theta_1' \theta_3' dx; T_5 = \int_0^l (\theta_3')^2 dx \int_0^l \theta_1' \theta_3' dx; T_6 = \int_0^l (\theta_3')^2 dx \int_0^l \theta_1' \theta_2' dx; \\
T_7 &= \int_0^l (f_1')^2 dx \int_0^l (\theta_3')^2 dx; T_8 = \int_0^l (\theta_3')^2 dx \int_0^l (f_2')^2 dx; T_9 = \int_0^l (\theta_3')^2 dx \int_0^l (f_3')^2 dx; T_{10} = \int_0^l (\theta_3')^2 dx \int_0^l f_1' f_2' dx; \\
T_{11} &= \int_0^l (\theta_3')^2 dx \int_0^l f_1' f_3' dx; T_{12} = \int_0^l (\theta_3')^2 dx \int_0^l f_2' f_3' dx; T_{14} = \int_0^l \theta_1^3 \theta_3 dx; T_{15} = \int_0^l \theta_2^3 \theta_3 dx; T_{16} = \int_0^l \theta_3^4 dx; \\
T_{17} &= \int_0^l \theta_1^2 \theta_2 \theta_3 dx; T_{18} = \int_0^l \theta_1^2 \theta_3^2 dx; T_{19} = \int_0^l \theta_1 \theta_2^2 \theta_3 dx; T_{20} = \int_0^l \theta_2^2 \theta_3^2 dx; T_{21} = \int_0^l \theta_2 \theta_3^3 dx; T_{22} = \int_0^l \theta_1 \theta_3^3 dx; \\
T_{23} &= \int_0^l \theta_1 \theta_2 \theta_3^2 dx; I_{ps} = \int_0^l \theta_3 \theta_0 dx;
\end{aligned}$$

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